

Research Paper

Impact of interfacial kinetics on Marangoni instabilities in a moderately dense nanofluid heated from below

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ARTICLE INFO

Keywords:

Nanofluid layer
Thermocapillary effects
Solutocapillary effects
Nanoparticle interfacial kinetics

ABSTRACT

As found in our previous work, a nanofluid layer with a free interface cooled at the substrate exhibits the emergence of both thermocapillary and solutocapillary instabilities driven by the stratification of thermophysical properties and the presence of interfacial kinetics of nanoparticles. In the present paper, a similar nanofluid layer heated at the substrate displays the emergence of the thermocapillary and solutocapillary instabilities driven by fundamentally different physical mechanisms. We investigate the onset of thermosolutal instabilities in a moderately dense nanofluid layer heated at the solid substrate and exposed to the ambient gas phase at its deformable liquid–gas interface in the gravity field and in the presence of thermophoresis and interfacial adsorption/desorption kinetics of nanoparticles. We note that the nanoparticle surface energetics (adsorption and desorption rates) promote an intrinsic non-monotonic variation in the surface tension with nanoparticle concentration, which is taken into account here. Linear stability analysis of the quiescent base state of the system is investigated using normal modes, and the resulting eigenvalue problem is solved numerically. We find that the change of the heating direction leads to a switch of the emerging instability from short-wave to long-wave. In the case of heating at the bottom, we reveal that the emerging thermo-solutocapillary instability is monotonic and long-wave. In this context, a wide range of long-wave scaling laws is determined.

1. Introduction

Multi-phase systems (liquid–liquid or liquid–gas) often invoke capillary or surface tension-driven phenomena due to interfacial inhomogeneity of scalar fields, such as temperature, concentration, and electric field, leading to the emergence of convective motion driven by surface tension variation known as the *Marangoni* effect (Levich, 1962; Levich and Krylov, 1969). When this effect is induced by temperature, concentration, and the strength of an electric field, it is explicitly referred to as thermocapillarity, solutocapillarity, and electrocapillarity, respectively, and the instability caused by the Marangoni effect is called the Marangoni instability (Scriven and Sternling, 1960; Takashima, 1981; Davis, 1987; Probstein, 1994; Joo, 1995). Marangoni instability is ubiquitous in nature, physiology, and our day-to-day industrial products (Manikantan and Squires, 2020; Temprano-Coletto and Stone, 2024), such as drying paint (Pearson, 1958), flukeprint of whale (Levy et al., 2011), formation of ripples on the skinned surface of chocolate pudding near the cup center (Probstein, 1994), embryo and tissue growth (Gsell et al., 2025), surfactant transport in lung (Grotberg et al., 1995), single crystal growth in microgravity environment (Schwabe et al., 1978), controlling the motion of surface waves (Erinin et al.,

2023), the stability of foams and emulsions (Chatzigiannakis et al., 2025; Sharma et al., 2025), and breakup of a thread (Hameed et al., 2008), to mention a few.

Marangoni instability is not only perceived to affect crucial transport processes, for example, the stability of foams and emulsions, ink-jet printing (Lohse, 2022), and distillation (Edwards et al., 1991; Lyubimova et al., 2025), but it is also intentionally used to enhance a transport process, such as capillary pumping of pharmaceutical components (Frumkin et al., 2014), suppression of Rayleigh–Taylor instability (Oron and Rosenau, 1992; Burgess et al., 2001; Alexeev and Oron, 2007), mixing in a low-gravity environment, bubble removal in a liquid glass, and prevention of radial dopant segregation in semiconductor crystal growth (Ostrach, 1982). Microfluidic devices, such as ‘lab-on-a-chip’, serve as an excellent platform to utilize surface tension-driven phenomena; for example, micrometer-sized droplet management in genetic analysis, prompt chemical detection technologies, and controlled droplet divisions (Davis, 1983; Oron et al., 1997; Schatz and Neitzel, 2001). As with technological advancements, complex systems and fluids are frequently encountered and employed. To begin with, surfactants, soluble or insoluble, are widely used for their ability to

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reduce surface tension at the fluid–fluid interface, leading to the emergence of solutocapillary flow (Palmer and Berg, 1972; Shklyaev and Nepomnyashchy, 2013; Morozov et al., 2014; Guzmán et al., 2022). There are exceptional cases in which the surface tension increases with concentration, such as binary mixtures containing an inorganic solute (Oron and Nepomnyashchy, 2004; Podolny et al., 2005; Shklyaev and Nepomnyashchy, 2017). For most liquids, surface tension decreases linearly with an increase in surface temperature (Probstein, 1994).

A colloidal dispersion consisting of a carrier fluid and nanoparticles of diameter $d_p^* \sim 10$ to 100 nm, known as a *nanofluid*, displays a non-trivial variation of surface tension with nanoparticle concentration due to the surface energetics of nanoparticles in a carrier liquid (Machrafi, 2022). Subsequently, depending on the nanoparticle interaction potential with the liquid–gas interface in a base liquid, nanoparticles may adsorb at the interface, and this may result in a non-monotonic variation of surface tension with particle concentration (Gandhi et al., 2026a). Moreover, the thermophysical properties of a nanofluid, such as density, viscosity, heat capacity, and thermal conductivity, depend strongly on the local particle concentration (Buongiorno, 2006). Nanofluids may also exhibit normal stress contributions to the total viscous stress (Phillips et al., 1992; Morris, 2020); however, these can be safely neglected in the absence of a base-state flow. Due to their specific thermophysical properties, nanofluids are engineered in a wide range of applications, for example, in heat transfer (Choi and Eastman, 1995), ink-jet printing (Lohse, 2022), paints (Uchiyama et al., 2015), microgravity experiments (Vailati et al., 2023), and nanofluid fuels (Abramzon and Sirignano, 1989; Basu and Miglani, 2016).

The theoretical framework for the investigation of dynamics and stability of fluids in the presence of soluble and insoluble surfactants or micelles (Craster and Matar, 2009; Manikantan and Squires, 2020) is well established. However, for nanofluids, a theoretical investigation taking into consideration interfacial kinetics of nanoparticles, the non-monotonicity of surface tension, and the stratification of thermophysical properties (Welander and Holm aker, 1971; Yih, 1980), requires further development. Additionally, the thermo-solutocapillary instability may emerge in short-wave or long-wave (Goldstone mode) range, depending on the system parameters. In the latter case, surface deformation plays a vital role by providing a slow modulation shift in the horizontal direction (Oron et al., 1997; Craster and Matar, 2009; Nepomnyashchy et al., 2001); however, in the former case, the impact of surface deformations is dominated by a strong capillary effect (Gandhi et al., 2026b,a).

Nanoparticles naturally tend to mutually attract and form clusters due to three major sources of interaction: van der Waals interactions, depletion interactions, and interactions arising from interfacial deformation (van Saarloos et al., 2023). Nevertheless, with technological advancements in colloidal science, structurally stable nanoparticle dispersions can be engineered by employing electrostatic stabilization (Russel, 1991; Russel et al., 1989) and steric stabilization by grafting polymers onto the surface of nanoparticles (Israelachvili, 2011; van Saarloos et al., 2023). While the former approach introduces an additional complexity to the system due to electrolyte concentration in the solvent and its impact on the thermophysical properties, the latter approach, remarkably, stabilizes the colloidal dispersion by grafting a thin soluble polymer layer of 2–5 nm thickness onto the nanoparticle surface and suppresses the interactions between the particles. On the other hand, the physicochemical properties of nanoparticles in the base liquid, surface charges, and topography may in general lead to irreversible nanoparticle adsorption at the solid substrate (Moore et al., 2025) or the adsorption/desorption kinetics at the solid–liquid interface (Zigelman and Manor, 2018). However, surface charges and topography of the latter can be tuned to suppress or weaken the nanoparticle adsorption at the solid substrate.

The mathematical model presented here follows our recent investigation (Gandhi et al., 2026b) in the setting of cooling at the substrate.

The novelty and key difference of this paper, as compared to our earlier investigation (Gandhi et al., 2025) are as follows: (i) it was assumed there that the equilibrium between the bulk and interfacial nanoparticle concentration is established infinitely fast, so that the processes of interfacial nanoparticle adsorption and desorption kinetics were ignored; (ii) it was also assumed there that the surface tension decreases with nanoparticle concentration, which is ideally applicable for relatively low nanoparticle concentrations. Subsequently, in Gandhi et al. (2026b,a), (i) and (ii) were reconsidered, taking into account interfacial nanoparticle kinetics, the dependence of surface tension on both bulk and surface nanoparticle concentrations, and, notably, an anomalous surface tension variation with the nanoparticle concentration. In particular, the findings presented in Gandhi et al. (2026b) concern thermosolutal instability in the presence of interfacial nanoparticle kinetics with cooling at the solid substrate and reveal the emergence of anomalous solutocapillary instability and the complex role of thermal conductivity stratification coupled with inhomogeneity of particle concentration. Naturally, we ask the following questions: what is in terms of the system stability the effect and consequences of heating a nanofluid layer at the solid substrate in the presence of interfacial kinetics and stratification of thermophysical properties? Will there be new instabilities due to the presence of stratification of the thermophysical properties of the nanofluid and interfacial kinetics? If so, will the instability emerge in the short- or long-wave range? These questions serve as a central motivation for this paper. A notable and direct implication of the investigation presented in the present paper is relevant, among other features, for the formation of colloidal crystals due to evaporation, foam and emulsion stability, to mention just a few here (Kralchevsky and Nagayama, 2001; Edwards et al., 1991). To emphasize further, by understanding and controlling the underlying physical mechanisms behind particle ordering, for example, in colloidal crystals, it would be possible to achieve improvements in a wide range of technological applications, such as optics for diffraction gratings and in the creation of nanostructured surfaces for catalytic films (Denkov et al., 1992).

The plan of the paper is as follows. Section 2 provides the general mathematical framework, including the dimensional and non-dimensional governing equations and boundary conditions, incorporating the particle concentration-dependent thermophysical properties of the nanofluid, and presents the quiescent base state of the system. Section 3 details the formulation of the linear stability analysis for the base state and the resulting eigenvalue problem. Section 4 is dedicated to the presentation of the results and to the pertinent discussion of these. In particular, Sections 4.1 and 4.2 reveal the emergence of long-wave pure thermal and solutal Marangoni instabilities, respectively. Section 4.3 briefly addresses the case of a combined thermo-solutocapillary instability. Finally, Section 5 concludes the present paper with a summary of the main results.

2. Problem formulation

We consider a three-dimensional nanofluid layer of a mean thickness h_0^* deposited on a rigid and impermeable horizontal substrate, and exposed at its deformable interface to the quiescent gas environment held at constant pressure p_∞^* and temperature T_∞^* in the gravity field g^* . A list of the thermophysical properties of the nanofluid includes the density ρ_{nf}^* , dynamic viscosity μ_{nf}^* , kinematic viscosity $\nu_{nf}^* = \mu_{nf}^*/\rho_{nf}^*$, thermal conductivity K_{nf}^* , heat capacity c_{nf}^* , and thermal diffusivity $\kappa_{nf}^* = K_{nf}^*/\rho_{nf}^*c_{nf}^*$, where the subscript nf denotes the properties of the nanofluid. In what follows, the properties of the base fluid and of nanoparticles are denoted with the same letters but decorated with the subscripts bf and np , respectively. The nanofluid layer is assumed to be heated at its solid bottom with the prescribed heat flux q^* , $q^* > 0$ in the vertical direction, which leads to the emergence of the temperature field $T^* = T^*(x^*, y^*, z^*, t^*)$ that varies with time and space within the layer.

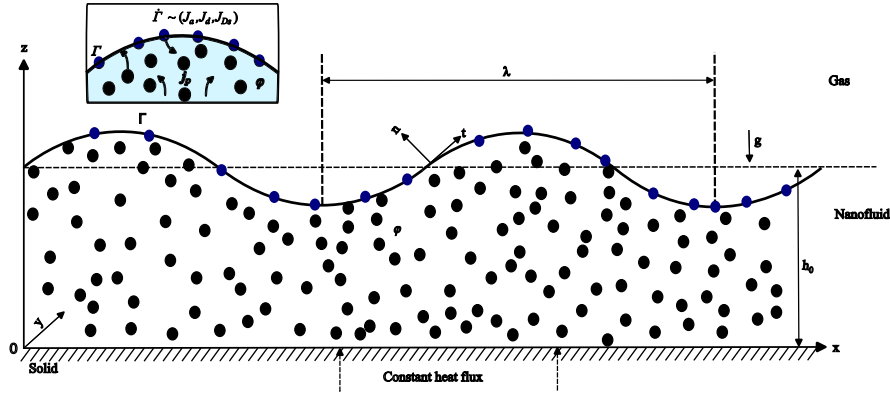


Fig. 1. Nanofluid layer with particles of diameter $d_p^* \sim 50$ nm deposited on the solid substrate, subjected to a constant heat flux at the substrate and exposed to the gas phase at its deformable interface with interfacial adsorption/desorption particle kinetics.

We choose the frame of reference such that the x^* and y^* axes are located on the substrate, whereas the z^* axis is normal to the substrate and directed into the fluid layer in the direction opposite to that of gravity. In this setting, the substrate and the deformable liquid–gas interface of the layer are located at $z^* = 0$ and $z^* = h^*(x^*, y^*, t^*)$ at time t^* , respectively, as shown in Fig. 1.

The nanofluid layer is assumed to be moderately dense in terms of the volumetric nanoparticle concentration $\phi^* = \phi^*(x^*, y^*, z^*, t^*)$, which varies in space (x^*, y^*, z^*) and time t^* . We emphasize that the notion of a ‘moderately dense’ nanofluid signifies that the thermophysical properties of the nanofluid are no longer constant, as in the case of a dilute suspension or a binary mixture. Instead, the local particle concentration is still low, $\phi^* < 0.1$, but it is sufficiently high so that the leading-order contribution $O(\phi^*)$ for the thermophysical properties of the nanofluid is important and cannot be ignored beforehand. The hard spherical Al_2O_3 (alumina) particles with a diameter $d_p^* \approx 50$ nm are freely suspended in a layer of a Newtonian carrier fluid. Furthermore, nanoparticles in a carrier liquid exhibit an intrinsic nature of interaction with the surrounding gas phase, which consequently leads to nanoparticle adsorption and desorption at/from the air–liquid interface (Machrafi, 2022). The processes of adsorption and desorption, respectively, reflect the presence of an energy barrier and a primary minimum in the interaction potential (Grow and Shaeiwitz, 1982) and their rates are derived from the latter. We denote the interfacial nanoparticle concentration by $\Gamma^*(x^*, y^*, t^*)$. Furthermore, lateral nanoparticle interactions at the interface (Adamczyk, 2012) are neglected; therefore, only the effects of vertical interfacial kinetics are considered. We also assume that the nanofluid represents here a structurally stable colloidal suspension of hard spheres with no interactions between them or between them and the solid substrate underneath.

2.1. Physical mechanisms related to non-isothermal nanofluid layer

The absolute velocity of a nanoparticle is described by the sum of the base fluid velocity and the relative (slip) velocity. Buongiorno (2006) described seven different slip mechanisms for a nanofluid suspension, which are inertia, Brownian diffusion, thermophoresis or the Soret effect, diffusiophoresis, Magnus effect, fluid drainage, and gravitational settling. Out of these seven, we consider the two dominant slip mechanisms, i.e., the Brownian diffusion and the Soret effect.

The Brownian diffusion coefficient valid for the arbitrary volume fraction of nanoparticles is given by the generalized Stokes-Einstein formula (Russel et al., 1989)

$$\mathcal{D}^*(\phi^*) = D_B (1 - \phi^*)^{2.55} (1 + 4\phi^* + 4\phi^{*2} - 4\phi^{*3} + \phi^{*4}), \quad (1)$$

where D_B is the Brownian diffusivity given by the Stokes-Einstein formula as

$$D_B = \frac{K_B^* T^*}{3\pi\mu_{bf}^* d_p^*}, \quad (2)$$

and K_B^* is the Boltzmann constant, $K_B^* = 1.380649 \times 10^{-23}$ J/K. In the limit of a moderate nanoparticle concentration ϕ^* , the generalized Stokes-Einstein formula (1) reduces to the linear model

$$\mathcal{D}^*(\phi^*) = D_B (1 + 1.45\phi^*) \quad (3)$$

derived by Batchelor (1976). We note that our earlier work (Gandhi et al., 2025) illustrates that the concentration-dependent Brownian diffusivity imparts a negligible contribution to the threshold of thermosolutal Marangoni instability; thence, for the sake of simplicity, we consider the constant Brownian diffusivity coefficient $\mathcal{D}^*(\phi^*) = D_B$. In a non-isothermal nanofluid layer, the temperature gradient between the solid substrate and the deformable interface created by the heat flux induces another slip mechanism known as *thermophoresis* or the Soret effect.

We express the total nanoparticle mass flux by a sum of the mass flux induced by Brownian diffusion and that due to the Soret effect:

$$\mathbf{j}_p = -\rho_{np}^* \left(C_B T^* \nabla^* \phi^* + C_T \phi^* \frac{\nabla^* T^*}{T^*} \right), \quad (4)$$

where $C_B = D_B/T^*$, $C_T = D_T/\phi^*$, and the Soret coefficient for the single-component nanofluid mixture is given by (Buongiorno, 2006)

$$D_T = \left(\frac{0.26 K_{bf}^*}{2 K_{bf}^* + K_{np}^*} \right) \frac{\mu_{bf}^*}{\rho_{bf}^*} \phi^*. \quad (5)$$

$\nabla^* = (\partial_{x^*}, \partial_{y^*}, \partial_{z^*})$ with subscripts x^*, y^*, z^* representing partial differentiation with respect to the corresponding variable. We note that the total mass flux of nanoparticles given by Eq. (4) ensures a positively definite distribution of nanoparticle concentration, even subjected to strong thermophoresis. The other five slip mechanisms are negligible for thin nanofluid layers with particles of diameter $d_p^* = 50$ nm. However, gravitational settling becomes important for thick nanofluid layers or in suspensions with larger nanoparticles (Gandhi et al., 2025).

We assume that the heat flux in the nanofluid is given by Fourier’s law of heat conduction

$$\mathbf{j}_T = -K_{nf}^* \nabla^* T^*, \quad (6)$$

where we neglect the heat transfer contribution due to the Dufour effect and the nanoparticle mass transfer. The Dufour effect is negligible for liquid mixtures (Cross and Hohenberg, 1993; Oron and Nepomnyashchy, 2004); however, it is important for heat transfer in gases. We note that other components (Buongiorno, 2006) of nanoparticle mass transfer can also contribute to overall heat transfer, for instance, gravitational sedimentation (Chang and Ruu, 2022). However, for a thin nanofluid layer with small-sized nanoparticles, they are negligible (Gandhi et al., 2025).

2.2. Governing equations and boundary conditions

The thermophysical properties of a moderately dense nanofluid, such as density, heat capacity, dynamic viscosity, and thermal conductivity, strongly depend on nanoparticle concentration (Buongiorno, 2006). A detailed discussion on the relevant thermophysical properties and a comparison with other models can be found in (Gandhi et al., 2025, 2026b).

The dynamics of a nanofluid layer is governed by the set of dimensional governing equations, which consists of the continuity, Navier–Stokes, advection-conduction heat transfer, and nanoparticle mass transfer equations, respectively (Bird et al., 2002; Colinet et al., 2001),

$$(\rho_{nf}^*)_{t^*} + \nabla^* \cdot (\rho_{nf}^* \mathbf{u}^*) = 0, \quad (7a)$$

$$(\rho_{nf}^* \mathbf{u}^*)_{t^*} + \nabla^* \cdot (\rho_{nf}^* \mathbf{u}^* \mathbf{u}^*) = -\nabla^* p^* + \nabla^* \cdot \boldsymbol{\tau}^* - \rho_{nf}^* g^* \mathbf{e}_{z^*}, \quad (7b)$$

$$(\rho^* c^*)_{nf} [T_{t^*}^* + (\mathbf{u}^* \cdot \nabla^*) T^*] = \nabla^* \cdot (K_{nf}^* \nabla^* T^*), \quad (7c)$$

$$\phi_{t^*}^* + \nabla^* \cdot (\mathbf{u}^* \phi^*) = \nabla^* \cdot \left(C_B T^* \nabla^* \phi^* + C_T \phi^* \frac{\nabla^* T^*}{T^*} \right), \quad (7d)$$

where g^* is the gravity acceleration, $\mathbf{e}_{z^*}$ is the unit base vector in the z^* direction, $\mathbf{u}^* = (u^*, v^*, w^*)$ is the flow velocity field vector, and $\boldsymbol{\tau}^*$ is the viscous part of the stress tensor $\boldsymbol{\tau}^* = \mu_{nf}^* [\nabla^* \mathbf{u}^* + (\nabla^* \mathbf{u}^*)^T] - \left(\frac{2}{3} \mu_{nf}^* - \mathcal{K}^* \right) (\nabla^* \cdot \mathbf{u}^*) \mathbf{I}$, where the superscript T denotes the transpose of the corresponding tensor, $\mathbf{I}$ is the unity tensor, the subscript t^* stands for a partial derivative with respect to time t^* and $\mathcal{K}^*$ is the dilatational viscosity of the fluid. We note that the last term in Eq. (7b) represents the buoyancy force, which is due to the fluid density varying within the layer with the particle concentration that is, in turn, coupled to the fluid temperature via the governing equations (7).

We impose three boundary conditions at the solid–liquid interface $z^* = 0$:

$$\mathbf{u}^* = 0, \quad -K_{nf}^* T_{z^*}^* = q^*, \quad C_B T^* \phi_{z^*}^* + C_T \phi^* \frac{T_{z^*}^*}{T^*} = 0. \quad (8)$$

The first boundary condition represents the no-slip and no-penetration for the flow velocity at the substrate. The second condition represents a specified heat flux q^* (Rohsenow et al., 1998; Platten and Legros, 1984) at the solid substrate, appropriate for a substrate made of a material with low thermal conductivity. Finally, the third condition corresponds to the impermeability of the substrate in terms of the total mass flux of the nanoparticles.

The interfacial motion of a deformable liquid–gas interface at $z^* = h^*(x^*, y^*, t^*)$ is governed by the following kinematic boundary condition (Gandhi et al., 2026b)

$$h_{t^*}^* + u^* h_{x^*}^* + v^* h_{y^*}^* = w^*. \quad (9a)$$

Furthermore, at a deformable liquid–gas interface, the continuity of total interfacial stress balance is valid

$$\mathbf{n}^* \cdot \mathcal{T}^* - \mathbf{n}^* \cdot \widehat{\mathcal{T}}^* = 2\mathcal{H}^* \mathbf{n}^* \sigma^* + \nabla_s^* \sigma^*, \quad (9b)$$

where $\mathcal{T}^* = -p^* \mathbf{I} + \boldsymbol{\tau}^*$ is the total stresses exerted by the nanofluid at the liquid–gas interface, whereas $\widehat{\mathcal{T}}^* = -p_g^* \mathbf{I} + \boldsymbol{\tau}_g^*$ represents the total stress contribution by the gas phase. We, however, neglect the total viscous stress exerted by the gas phase, under the assumption of a small viscosity ratio ≈ 0.02 (Colinet et al., 2001; Nepomnyashchy et al., 2001). It follows that the total stress balance at the deformable interface (9b) yields the normal and tangential stress balance at $z^* = h^*(x^*, y^*, t^*)$

$$\begin{aligned} \mathbf{n}^* \cdot \mathcal{T}^* \cdot \mathbf{n}^* &= 2\mathcal{H}^* \sigma^*, & \mathbf{n}^* \cdot \mathcal{T}^* \cdot \mathbf{x}^* &= \mathbf{x}^* \cdot \nabla_s^* \sigma^*, \\ \mathbf{n}^* \cdot \mathcal{T}^* \cdot \mathbf{y}^* &= \mathbf{y}^* \cdot \nabla_s^* \sigma^*, \end{aligned} \quad (9c)$$

where we follow the Stokes' hypothesis (Buresti, 2015) and assume the zero value of the dilatational viscosity $\mathcal{K}^*$. This implies that the

absolute value of $\mathcal{K}^* \nabla^* \cdot \mathbf{u}^*$ is negligible compared to the thermodynamic pressure p^* , i.e., $|\mathcal{K}^* \nabla^* \cdot \mathbf{u}^*| \ll p^*$. We note that the value of the surface tension σ^* will be derived below in Eq. (10). Here, we express the surface gradient operator $\nabla_s^* = (\mathbf{I} - \mathbf{n}^* \mathbf{n}^*) \cdot \nabla^*$, and unit vector normal via

$$\mathbf{n}^* = \frac{-h_{x^*}^* \mathbf{e}_{x^*} - h_{y^*}^* \mathbf{e}_{y^*} + \mathbf{e}_{z^*}}{\sqrt{1 + h_{x^*}^{*2} + h_{y^*}^{*2}}}, \quad (9d)$$

the unit tangent vectors to the deformable liquid–gas interface by

$$\mathbf{x}^* = \frac{\mathbf{e}_{x^*} + h_{x^*}^* \mathbf{e}_{z^*}}{\sqrt{1 + h_{x^*}^{*2}}}, \quad \mathbf{y}^* = \frac{\mathbf{e}_{y^*} + h_{y^*}^* \mathbf{e}_{z^*}}{\sqrt{1 + h_{y^*}^{*2}}}, \quad (9e)$$

and the mean interfacial curvature through $2\mathcal{H}^* = -\nabla_s^* \cdot \mathbf{n}^*$. Furthermore, we denote the unit base orthonormal vectors in x^* and y^* by $\mathbf{e}_{x^*}$ and $\mathbf{e}_{y^*}$, respectively.

In a non-isothermal nanofluid layer, the heat transfer at the interface $z^* = h^*(x^*, y^*, t^*)$ is expressed by Newton's law of cooling

$$-K_{nf}^* \nabla^* T^* \cdot \mathbf{n}^* = \hat{q}(T^* - T_\infty^*), \quad (9f)$$

where $\hat{q}$ is the rate of heat transfer from liquid to gas phase by convection. Note that the heat transfer boundary condition effectively couples the temperature and local particle concentration at the interface since K_{nf}^* depends on the latter.

Additionally, the interfacial concentration Γ^* , in units of mol/m², is expressed by (Machrafi, 2022)

$$\Gamma^* = \vartheta \Gamma_\infty^*, \quad (9g)$$

where ϑ and $\Gamma_\infty^* = \vartheta_\infty \mathcal{L}_{np}^* \frac{\rho_{np}^*}{\mathcal{M}_{np}^*}$ are the surface coverage and the maximum surface concentration, respectively. For spherical particles, the maximum geometrical coverage fraction $\vartheta_\infty = 0.547$, the characteristic length $\mathcal{L}_{np}^* = d_p^*/3$, and the factor

$$\frac{\rho_{np}^*}{\mathcal{M}_{np}^*} = \left[\frac{4\pi^2}{9\sqrt{2}} \left(\frac{d_p^*}{2} \right)^3 N_A^* \right]^{-1} \quad (9h)$$

yields an equivalent bulk nanoparticle concentration volume fraction to the interfacial molar mass, where $N_A^* = 6.02 \times 10^{23} \text{ mol}^{-1}$ is the Avogadro's number. The molar mass of a nanoparticle is given by (Machrafi, 2022)

$$\mathcal{M}_{np}^* = \mathcal{M}_{np}^{*t} f_{np} \frac{\mathcal{V}_{np}^*}{\mathcal{V}_{np}^{*t}}, \quad (9i)$$

where $\mathcal{M}_{np}^{*t}$ and $f_{np} = \frac{\pi}{3\sqrt{2}}$ denote the molar mass of one molecule and maximum 3-dimensional nanoparticle packing density, e.g., face-centered cubic (fcc) of spheres, respectively. It follows that the volumes of a nanoparticle and a spherical molecule are defined, respectively, as:

$$\mathcal{V}_{np}^* = \frac{4\pi}{3} \left(\frac{d_p^*}{2} \right)^3 \quad \text{and} \quad \mathcal{V}_{np}^{*t} = \frac{4\pi}{3} \ell_{np}^{*3}, \quad \ell_{np}^* = \left(\frac{3\mathcal{M}_{np}^{*t}}{4\pi \rho_{np}^* N_A^*} \right)^{1/3}. \quad (9j)$$

Here, ℓ_{np}^* depicts the radius of a molecule; similarly, we will define the effective radius of a base fluid molecule $\mathcal{L}_{bf}^*$.

At the liquid–gas deformable interface $z^* = h^*(x^*, y^*, t^*)$, the interfacial nanoparticle concentration varies in time and space. The evolution of adsorbed interfacial nanoparticle concentration is given by the extension of the conservation equation derived for the case without interfacial kinetics $\mathcal{J}_a^* = \mathcal{J}_d^* = 0$ (Scriven, 1960; Wong et al., 1996; Manikantan and Squires, 2020; Edwards et al., 1991)

$$\begin{aligned} \frac{\partial \Gamma^*}{\partial t^*} - \frac{\partial h^*}{\partial t^*} (\mathbf{e}_{z^*} \nabla_s^* \Gamma^*) + \nabla_s^* \cdot (\mathbf{u}_s^* \Gamma^*) \\ + \Gamma^* (\nabla_s^* \cdot \mathbf{n}^*) (\mathbf{u}^* \cdot \mathbf{n}^*) = -\nabla_s^* \mathcal{J}_{Ds}^* + \Gamma_\infty^* \mathcal{J}_a^* [\mathcal{J}_a^* - \mathcal{J}_d^*], \end{aligned} \quad (9k)$$

where $\mathbf{u}_s^* = (\mathbf{I} - \mathbf{n}^*\mathbf{n}^*) \cdot \mathbf{u}^*$ is the surface velocity component and the specific surface area covered by the interfacial nanoparticles per their unit mole $\mathcal{S}_a^*$ is given by

$$\mathcal{S}_a^* = \frac{\mathcal{M}_{np}^*}{\rho_{np}^* \mathcal{L}_{np}^*}. \quad (9l)$$

The evolution equation (9k) introduces fluxes due to surface advection $\mathbf{u}_s^* \Gamma^*$, surface diffusion along the interface $\mathcal{J}_{Ds}^*$, and for the soluble interfacial nanoparticles, the adsorption flux $\mathcal{J}_a^*$ and the desorption flux $\mathcal{J}_d^*$. We express the surface diffusion flux via Fick's law of diffusion with a constant surface diffusivity D_s^*

$$\mathcal{J}_{Ds}^* = -D_s^* \nabla_s^* \Gamma^*. \quad (9m)$$

We note that for the surfactant solutions, similarly to the bulk concentration, the surface diffusivity generally displays the dependence on the interfacial concentration (Manikantan and Squires, 2020; Valkovska and Danov, 2000), analogous to the bulk diffusion coefficient $\mathcal{D}^*(\phi^*)$ given by Eq. (3). However, the variation of surface diffusion in the domain of moderate nanoparticle concentration is quite small. Thus, in what follows, we proceed with the constant surface diffusivity.

The adsorption/desorption flux characterizes the nanoparticle exchange between the bulk and the air-liquid interface. We adopt the Langmuir isotherm model for a moderately dense colloidal suspension. The adsorption flux $\mathcal{J}_a^*$ is thus expressed by the adsorption of the bulk molar concentration C^* to the available empty interfacial space $(1 - \Gamma^*/\Gamma_\infty^*)$ with the adsorption rate k_a^* , hence, the adsorption flux is given by

$$\mathcal{J}_a^* = k_a^* C^* \left(1 - \frac{\Gamma^*}{\Gamma_\infty^*}\right), \quad (9n)$$

where $C^* = \frac{\rho_{np}^*}{\mathcal{M}_{np}^*} \phi^*$ is the bulk molar nanoparticle concentration. The desorption flux $\mathcal{J}_d^*$ of the adsorbed nanoparticles per specific area with a desorption rate k_d^* is expressed by

$$\mathcal{J}_d^* = \frac{k_d^*}{\mathcal{S}_a^*} \left(\frac{\Gamma^*}{\Gamma_\infty^*}\right). \quad (9o)$$

We subsequently substitute Eqs. (9m), (9n) and (9o) into Eq. (9k), and deduce the generalized interfacial spatiotemporal evolution equation for the interfacial nanoparticle concentration.

At the deformable interface, the total normal molar mass flux balances the total interfacial kinetic flux

$$-\left(C_B T^* \nabla^* C^* + C_T C^* \frac{\nabla^* T^*}{T^*}\right) \cdot \mathbf{n}^* = \mathcal{J}_a^* - \mathcal{J}_d^*. \quad (9p)$$

In a nanofluid layer with heat and mass transfer, the surface tension at the liquid-gas interface exhibits variation with local interfacial temperature and nanoparticle concentration. The variation of surface tension σ^* with temperature T^* is assumed to be linear when the particle concentration at the interface is kept constant $\sigma^* = \sigma_r^* - \sigma_{T^*}^* (T^* - T_r^*)$, with $\sigma_{T^*}^* \equiv -\partial\sigma^*/\partial T^* > 0$ and the subscript r denotes some reference values. The equation of state for the variation of surface tension with nanoparticle concentration, following the Gibbs adsorption isotherm (Gandhi et al., 2026a), which depends on both the interfacial and bulk nanoparticle concentrations, is given next.

The generalized surface tension model for a nanofluid with heat and mass transfer derived in Gandhi et al. (2026a) reads

$$\begin{aligned} \sigma^*(T^*, \Gamma^*, \phi^*) &= \sigma_r^* - \sigma_{T^*}^* (T^* - T_\infty^*) \\ &+ T_\infty^* N_A^* \Gamma_b^* K_B^* \left\{ -K_\Sigma \frac{\Gamma^*}{\Gamma_\infty^*} + \phi^* \ln \left(\frac{\Gamma^*}{\Gamma_\infty^*} \right) + \right. \\ &\left. \omega_{np} \left[(K_\Sigma - \phi^*) \ln \left(1 - \frac{\Gamma^*}{\Gamma_\infty^*} \right) + K_\Sigma \frac{\Gamma^*}{\Gamma_\infty^*} \right] \right\}, \end{aligned} \quad (10)$$

where σ_r^* is the equilibrium reference value of the base fluid surface tension with the gas, $\sigma_{T^*}^* \equiv -\partial\sigma^*/\partial T^* > 0$ expresses the variation of surface tension with temperature,

$$\omega_{np} = \frac{(\rho_{bf}^* / \mathcal{M}_{bf}^*) \mathcal{L}_{bf}^*}{(\rho_{np}^* / \mathcal{M}_{np}^*) \mathcal{L}_{np}^*} \quad (11)$$

represents the ratio between the specific interfacial area per mole occupied by nanoparticles to that covered by the base-fluid molecules, $\mathcal{M}_{bf}^*$ is the molar mass of the base fluid, $\mathcal{M}_{np}^*$ is the molar mass of the nanoparticle given by Eq. (9i), and

$$\mathcal{L}_{bf}^* = \frac{1}{3} \left(\frac{3 \mathcal{M}_{bf}^*}{4 \pi \rho_{bf}^* N_A^*} \right)^{1/3}, \quad \mathcal{L}_{np}^* = \frac{d_p^*}{3} \quad (12)$$

are the effective radii of the base fluid molecule and the nanoparticle, respectively. The parameters $\Gamma_b^* = \frac{\mathcal{L}_{np}^{*2} \rho_{np}^*}{\mathcal{L}_{bf}^* \mathcal{M}_{np}^*}$ and $K_\Sigma = \frac{\Gamma_\infty^*}{\Gamma_b^*}$ reflects the ability of nanoparticles to adsorb to the interface or stay in the bulk, and $N_A^* = 6.02 \times 10^{23} \text{ mol}^{-1}$ is the Avogadro's number.

We recall that the surface tension usually decreases with temperature and its variation is fairly well approximated by a linear function of temperature over small ranges of temperature variation. Thus the surface tension gradient with respect to temperature $\sigma_{T^*}^* = -\partial\sigma^*/\partial T^*$ is a positive quantity. However, we note that the equation of state for the surface tension given by Eq. (10) varies with both the bulk ϕ^* and interfacial nanoparticle concentrations Γ^* . Surface tension may exhibit a nonmonotonic variation with the nanoparticle concentration; notably, its character can vary with the properties of the carrier liquid affecting the interfacial nanoparticle kinetics (Machrafi, 2022; Gandhi et al., 2026b,a).

At last, the conservation of the total molar mass of nanoparticles $\frac{\rho_{np}^*}{\mathcal{M}_{np}^*} \Phi^*$ with Φ^* being the average volume concentration of the particles is imposed. It reads

$$\left(\frac{\rho_{np}^*}{\mathcal{M}_{np}^*} \right) I(\mathcal{D}^*) h_0^* \Phi^* = \int_0^{h_0^*} \left(\iint_{\mathcal{D}^*} \frac{\rho_{np}^*}{\mathcal{M}_{np}^*} \phi^*(x^*, y^*, z^*, t^*) dx^* dy^* \right) dz^* + \iint_{\mathcal{L}^*} \Gamma^*(x^*, y^*, t^*) dS^*, \quad (13a)$$

with

$$dS^* = \sqrt{1 + h_{x^*}^{*2} + h_{y^*}^{*2}} dx^* dy^*, \quad (13b)$$

where $\mathcal{D}^*$ is a projection of the flow domain onto the $x^* - y^*$ plane and $I(\mathcal{D}^*)$ is the area of this projection, whereas $\mathcal{L}^*$ represents the projection of the gas-liquid interface onto the $x^* - y^*$ surface. In the left-hand side of Eq. (13a), the average particle concentration Φ^* is multiplied by the domain volume, namely $I(\mathcal{D}^*) h_0^*$, to obtain the total molar mass of the nanoparticles in the flow bulk and at the interface of a nanofluid layer (Edmonstone et al., 2006). Eq. (13a) provides a needed closure of the problem by specifying a fixed value of Φ^* . We remind that the condition of the fixed nanoparticle average bulk concentration Φ^* is appropriate for the case of the quiescent base state. However, in the presence of a base state flow, the conservation of the nanoparticle mass flux should be employed (Ramachandran and Leighton, 2008; Morris, 2020).

2.3. Dimensionless formulation

The dimensional set of governing equations and boundary conditions given by Eqs. (7), (8) and (9) is brought to nondimensional form using the following scales:

$$\begin{aligned} (x^*, y^*, z^*) &= h_0^* (x, y, z), \quad h^* = h_0^* h, \quad \mathbf{u}^* = \frac{\kappa_{bf}^*}{h_0^*} \mathbf{u}, \quad t^* = \frac{h_0^{*2}}{v_{bf}^*} t, \\ \phi^* &= \phi_m \phi, \quad \sigma^* = \frac{\mu_{bf}^* \kappa_{bf}^*}{h_0^*} \sigma, \end{aligned}$$

$$p^* = p_\infty^* + \frac{\mu_{bf}^* \kappa_{bf}^*}{h_0^{*2}} p, \quad T^* = T_\infty^* + \frac{q^* h_0^*}{K_{bf}^*} T, \quad \Gamma^* = \Gamma_\infty^* \Gamma, \quad \Phi^* = \phi_m \Phi, \\ K_{nf}^* = K_{bf}^* K_{nf}, \quad \rho_{nf}^* = \rho_{bf}^* \rho_{nf}, \quad \mu_{nf}^* = \mu_{bf}^* \mu_{nf}, \\ (\rho^* c^*)_{nf} = (\rho^* c^*)_{bf} (\rho c)_{nf}. \quad (14)$$

Hereafter, the quantities without an asterisk represent the dimensionless variables. We obtain the following dimensionless governing equations:

$$P(\rho_{nf})_t + \nabla \cdot (\rho_{nf} \mathbf{u}) = 0, \quad (15a)$$

$$(\rho_{nf} \mathbf{u})_t + P^{-1} \nabla \cdot (\rho_{nf} \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} - \rho_{nf} G \mathbf{e}_z, \quad (15b)$$

$$(\rho c)_{nf} [PT_t + (\mathbf{u} \cdot \nabla)T] = \nabla \cdot (K_{nf} \nabla T), \quad (15c)$$

$$PL^{-1} \phi_t + L^{-1} \nabla \cdot (\mathbf{u} \phi) = \nabla^2 \phi + \nabla \cdot (\eta \phi \nabla T), \quad (15d)$$

where $\nabla \equiv \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$ and $\boldsymbol{\tau} = \mu_{nf} [\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} (\nabla \cdot \mathbf{u}) \mathbf{I}]$ is the viscous part of the dimensionless stress tensor.

The dimensionless boundary conditions at the solid substrate $z = 0$ are

$$\mathbf{u} = 0, \quad K_{nf} T_z = -1, \quad \phi_z + \eta \phi T_z = 0. \quad (16a)$$

At the deformable interface $z = h(x, y, t)$, the kinematic boundary condition and the continuity of normal and tangential stress balance are

$$P h_t + u h_x + v h_y = w, \quad (16b)$$

$$\mathbf{n} \cdot \mathcal{T} \cdot \mathbf{n} = 2\mathcal{H}\sigma, \quad \mathbf{n} \cdot \mathcal{T} \cdot \mathbf{x} = \mathbf{x} \cdot \nabla_s \sigma, \quad \mathbf{n} \cdot \mathcal{T} \cdot \mathbf{y} = \mathbf{y} \cdot \nabla_s \sigma, \quad (16c)$$

where the dimensionless total stress tensor is $\mathcal{T} = -p\mathbf{I} + \boldsymbol{\tau}$, and the normal and tangential vectors to the interface are

$$\mathbf{n} = \frac{-h_x \mathbf{e}_x - h_y \mathbf{e}_y + \mathbf{e}_z}{\sqrt{1 + h_x^2 + h_y^2}}, \quad \mathbf{x} = \frac{\mathbf{e}_x + h_x \mathbf{e}_z}{\sqrt{1 + h_x^2}}, \quad \mathbf{y} = \frac{\mathbf{e}_y + h_y \mathbf{e}_z}{\sqrt{1 + h_y^2}}, \quad \text{with} \\ 2\mathcal{H} = -\nabla_s \cdot \mathbf{n}, \quad (16d)$$

wherein, following the generalized surface tension model (10), we express the dimensionless equation of state with heat and mass transfer, as

$$\sigma = \Sigma_0 - M_T T + E \left[\omega_{np} [(K_\Sigma - \phi \phi_m) \ln(1 - \Gamma) + K_\Sigma \Gamma] - K_\Sigma \Gamma + \phi \phi_m \ln(\Gamma) \right]. \quad (16e)$$

The dimensionless heat transfer boundary condition at the deformable interface is

$$K_{nf} \nabla T \cdot \mathbf{n} + B T = 0. \quad (16f)$$

Finally, the dimensionless evolution equation for the interfacial nanoparticle concentration and the normal bulk nanoparticle mass flux balance with the interfacial kinetic flux at the liquid–gas interface are, respectively, given by

$$P [\Gamma_t - h_t (\mathbf{e}_z \cdot \nabla_s) \Gamma] + \nabla_s \cdot (\mathbf{u}_s \Gamma) + \Gamma (\nabla_s \cdot \mathbf{n}) (\mathbf{u} \cdot \mathbf{n}) = L_S (\nabla_s^2 \Gamma) + \mathcal{B}_A [K_{ad} \phi (1 - \Gamma) - \Gamma], \quad (16g)$$

$$-(\nabla \phi + \eta \phi \nabla T) \cdot \mathbf{n} = \mathcal{B}_D [K_{ad} \phi (1 - \Gamma) - \Gamma]. \quad (16h)$$

We note that the given dimensionless problem possesses a long list of dimensionless numbers governing the dynamics of the system via the following dimensionless numbers,

$$P = \frac{\nu_{bf}^* \rho_{bf}^* c_{bf}^*}{K_{bf}^*}, \quad G = \frac{g^* h_0^{*3}}{\nu_{bf}^* \kappa_{bf}^*}, \quad L = \frac{C_B T_\infty^* \rho_{bf}^* c_{bf}^*}{K_{bf}^*}, \quad L_s = \frac{D_s^* \rho_{bf}^* c_{bf}^*}{K_{bf}^*},$$

$$B = \frac{\hat{q} h_0^*}{K_{bf}^*}, \quad \Sigma_0 = \frac{\sigma^* h_0^*}{\mu_{bf}^* \kappa_{bf}^*}, \quad \eta = \frac{C_T q^* h_0^*}{C_B T_\infty^{*2} K_{bf}^*}, \quad (17a)$$

representing Prandtl, the modified Galileo, Lewis, surface Lewis, Biot, dimensionless surface tension numbers, and the Soret coefficient, respectively. The convective heat transfer induces the interfacial surface tension gradient, which is quantified by the dimensionless thermal Marangoni number

$$M_T = \frac{\sigma_{T^*}^* \Delta T^* h_0^*}{\mu_{bf}^* \kappa_{bf}^*}, \quad (17b)$$

where

$$\Delta T^* \equiv \frac{q^* h_0^*}{K_{bf}^*}. \quad (17c)$$

At the liquid–gas interface, the interfacial adsorption/desorption kinetics are controlled by the following dimensionless parameters

$$K_{ad} = \frac{k_a^* \rho_{np}^* \phi_m^* \mathcal{S}_a^*}{\mathcal{M}_{np}^* \kappa_d^*}, \quad \mathcal{B}_D = \frac{k_d^* h_0^* \mathcal{M}_{np}^*}{C_B T_\infty^* \phi_m^* \mathcal{S}_a^* \rho_{np}^*}, \quad \mathcal{B}_A = \frac{k_d^* h_0^{*2}}{\kappa_{bf}^*}, \quad (17d)$$

where the dimensionless number K_{ad} describes the interfacial adsorption/desorption kinetics through the ratio of the adsorption to desorption rate. The parameters $\mathcal{B}_D$ and $\mathcal{B}_A$ represent the ratio of the desorption flux to the diffusion flux and to the advection flux, respectively, whose ratio yields the dimensionless Péclet number $Pe = \mathcal{B}_D / \mathcal{B}_A$.

We define the dimensionless Elasticity number

$$E = \frac{h_0^* K_B^* N_A^* T_\infty^* \Gamma_b^*}{\mu_{bf}^* \kappa_{bf}^*}, \quad (18)$$

which describes the ratio of surface elasticity moduli at a fixed Γ_b^* to the dissipative factors (Seiwert et al., 2014; Manikantan and Squires, 2020). Note that according to Eq. (16e), the Elasticity number intrinsically quantifies the change in surface tension with interfacial nanoparticle concentration $\partial \sigma / \partial \Gamma$ and with bulk nanoparticle concentration $\partial \sigma / \partial \phi$. In fact, we notice the presence of two dynamic solutal Marangoni numbers (Gandhi et al., 2026b,a) (more precisely, Marangoni functions), namely, the surface concentration Marangoni number

$$M_\Gamma(\Gamma, \phi) \equiv \frac{\partial \sigma}{\partial \Gamma} = \frac{E(1 - \Gamma + \Gamma \omega_{np})(\Gamma K_\Sigma - \phi \phi_m)}{(\Gamma - 1)\Gamma}, \quad (19a)$$

and the bulk concentration Marangoni number

$$M_\phi(\Gamma) \equiv \frac{\partial \sigma}{\partial \phi} = E \phi_m [\ln(\Gamma) - \omega_{np} \ln(1 - \Gamma)]. \quad (19b)$$

We observe that the dynamic Marangoni numbers (19) contain the Elasticity number E , implying that the variation in E reflects the impact of the variation of the surface tension with the bulk and interfacial concentrations of particles under non-equilibrium conditions. The value of E represents an eigenvalue of the linear eigenvalue problem (EVP) formulated below and is associated with the possible onset of the solutocapillary instability. Furthermore, we define the nondimensional thermophysical properties that appear in the set of nondimensional governing equations and boundary conditions Eqs. (15) and (16) as

$$\rho_{nf}(\phi) = (1 - \phi \phi_m) + \phi \phi_m \rho_{np}, \quad (\rho c)_{nf}(\phi) = (1 - \phi \phi_m) + \phi \phi_m (\rho c)_{np}, \\ K_{nf}(\phi) = 1 + a \phi \phi_m, \quad \mu_{nf}(\phi) = (1 - \phi)^{-2}. \quad (20)$$

Finally, the dimensionless form of the conservation of the total particle mass across the nanofluid layer is given by

$$I(\mathcal{D})\Phi = \int_0^1 \left(\iint_{\mathcal{D}} \phi(x, y, z, t) dx dy \right) dz + \mathcal{A} \iint_{\mathcal{L}} \Gamma(x, y, t) dS, \quad (21a)$$

with

$$dS = \sqrt{1 + h_x^2 + h_y^2} dx dy, \quad (21b)$$

where $\mathcal{A} = \frac{\mathcal{M}_{np}^* \Gamma_\infty^*}{\rho_{np}^* h_0^* \phi_m^*}$ represents the particle adsorption factor at the liquid–gas interface.

2.4. Base state solution

We now search for a z -dependent steady quiescent ($\mathbf{u} = \mathbf{u}_0 = 0$) state of the system with the flat interface $z = 1$ in terms of the components of temperature $T_0(z)$, bulk nanoparticle concentration $\phi_0(z)$, pressure $p_0(z)$, and interfacial particle concentration Γ_0 . It follows from Eqs. (15b), (15a) and (15c) and the boundary conditions Eq. (16) that the steady state solution is determined by solving the following set of equations:

$$p_{0,z} = -G - G\phi_m(\rho_{np} - 1)\phi_0, \quad (22a)$$

$$[(1 + a\phi_m\phi_0)T_{0,z}]_z = 0, \quad (22b)$$

$$(\phi_{0,z} + \eta\phi_0T_{0,z})_z = 0, \quad (22c)$$

subject to the following boundary conditions:

$$(1 + a\phi_m\phi_0)T_{0,z} = -1, \quad \phi_{0,z} + \eta\phi_0T_{0,z} = 0 \quad \text{at } z = 0, \quad (23a)$$

$$p_0 = 0, \quad (1 + a\phi_m\phi_0)T_{0,z} + BT_0 = 0, \quad \phi_{0,z} + \eta\phi_0T_{0,z} = 0 \quad \text{at } z = 1. \quad (23b)$$

The thermodynamic equilibrium is characterized by minimization of the total change in Gibbs free energy, which accounts for the free energy cost to adsorb one particle to the interface and the free energy released due to the dissolution of the particle from the interface to the bulk phase. Hence, at equilibrium, the interfacial nanoparticle concentration is adequately described by the standard Langmuir isotherm (Machrafi, 2022; Manikantan and Squires, 2020). However, the value of the equilibrium adsorption constant there is purely empirical and is scarcely available for colloidal suspensions. Nevertheless, in the absence of empirical thermodynamic quantities, it is possible to find the equilibrium condition via dynamic equilibrium, i.e., the ratio of adsorption to desorption rate, instead of using the free energy (Machrafi, 2022).

Therefore, at equilibrium, the adsorption and desorption kinetic fluxes balance each other in Eq. (16h)

$$K_{ad}\phi_0(1 - \Gamma_0) - \Gamma_0 = 0. \quad (23c)$$

The base state of the system satisfying Eqs. (22) with the boundary conditions Eqs. (23) reads

$$T_0 = \frac{1}{\eta B} \left[B \ln \left(\frac{1}{a\phi_m} \mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta)) \right) - B \ln \left(\frac{1}{a\phi_m} \mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta z)) \right) + \eta \right], \quad (24a)$$

$$\phi_0 = \frac{\mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta z))}{a\phi_m}, \quad (24b)$$

$$\Gamma_0 = \frac{K_{ad}\mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta))}{K_{ad}\mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta)) + a\phi_m}, \quad (24c)$$

$$p_0 = G(1 - z) + \frac{1}{2\eta a} \left\{ G(\rho_{np} - 1) \left[\mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta)) - \mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta z)) \right] \times \left[\mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta)) + \mathcal{W}(\gamma(\Phi)a\phi_m \exp(\gamma(\Phi)a\phi_m + \eta z)) + 2 \right] \right\}, \quad (24d)$$

where $\gamma(\Phi)$ is a constant of integration to be determined below and $\mathcal{W}(z)$ is the principal branch of the Lambert W function (Corless et al., 1996; Gandhi et al., 2025) defined by

$$z = \mathcal{W}(z) \exp \mathcal{W}(z). \quad (25)$$

The details of the solution can be found in Gandhi et al. (2025).

Finally, we impose the conservation of particle volume in the bulk and at the interface by specifying the constant value of the average volume concentration of the nanoparticles Φ

$$\Phi = \int_0^1 \phi_0(z) dz + \mathcal{A}\Gamma_0, \quad (26)$$

whereby, we numerically obtain the value of $\gamma(\Phi)$ by substituting Eqs. (24b) and (24c) into Eq. (26). We note that the value of integration constant $\gamma(\Phi)$ depends on the system parameters a , ϕ_m , η , K_{ad} , and $\rho_{np}^*/\mathcal{M}_{np}^*$.

Fig. 2 illustrates the base state solution in terms of bulk nanoparticle concentration $\phi_0(z)$ in Fig. 2(a), and temperature $T_0(z)$ in Fig. 2(b), at different values of temperature difference ΔT^* across the nanofluid layer. We note that the Soret effect with positive values of the Soret coefficient drives the nanoparticles from a high-temperature solid substrate at $z = 0$ to the liquid–gas interface at $z = 1$, as shown in Fig. 2(a). We observe that the particle concentration of the base state exhibits a curved distribution with z across the layer height, and the curvature becomes more pronounced as the temperature difference ΔT^* increases. As a consequence, the nanoparticle distribution promotes the emergence of an unstable density stratification resulting in the maximal concentration at the interface.

We recall that in the case of cooling at the substrate, the concentration distribution illustrates a stable density stratification, which subsequently leads to the possible emergence of solutocapillary instability (Gandhi et al., 2025, 2026b). However, in the present setting, with heating at the solid substrate, the concentration profile is susceptible to the emergence of solutal buoyancy instability under the action of gravity when the nanoparticles are heavier than the carrier fluid, which is akin to the Rayleigh–Taylor instability (Chandrasekhar, 1961; Gandhi et al., 2025). The inset to Fig. 2(a) illustrates an increase in the equilibrium surface concentration Γ_0 with the Soret coefficient η . Since the Soret effect brings more particles towards the interface, in turn, the surface adsorption kinetics leads to a further increase in the surface particle concentration with η . However, the steady-state temperature $T_0(z)$ exhibits an almost linear variation with z across the layer with the maximal temperature attained on the substrate and decreases towards the interface, as illustrated in Fig. 2(b). We also find that the non-dimensional base state temperature remains almost independent of the temperature drop ΔT^* , as shown in Fig. 2(b).

Importantly, the investigation presented here deals with layers of a moderately dense nanofluid, which may be formally defined by

$$\Phi < 0.1 \text{ and } \phi_0(z) < 0.1 \quad (27)$$

for all $0 \leq z \leq 1$, which limits the particle content in terms of both the average and local volume concentration to 10% of the total volume of the nanofluid. The upper bounds for the average and local nanoparticle concentrations represent the essence of the moderately dense nanoparticle model, thus supporting the need in the $O(\phi)$ correction to the values of the thermophysical properties of the nanofluid. As shown in Fig. 2(a), the Soret effect leads to the emergence of a higher concentration of nanoparticles at the liquid–gas interface. In this situation, the local nanoparticle concentration there $\phi_0(1)$ is more than three times higher than Φ for the given parameter set. Note that the value of the quantity $\mathcal{A}\Gamma_0$ representing the interfacial nanoparticle concentration is explicitly determined, as shown in Fig. 2(a). We observe that this value is significantly lower than the subphase nanoparticle concentration, due to the lower particle solubility at the liquid–gas interface.

In turn, to rule out the emergence of high nanoparticle concentrations near the interface $z = 1$, we have to set a specific limitation, hereafter referred to as (i), for our theory to be valid. Another limitation, hereafter referred to as (ii), should be set for the upper bound of the temperature drop ΔT^* across the layer to exclude the temperature sensitivity of the thermophysical properties of the nanofluid. To first address limitation (ii), we assume that the temperature drop across the layer ΔT^* should not exceed 10 K, thus limiting the heat flux q^* at the

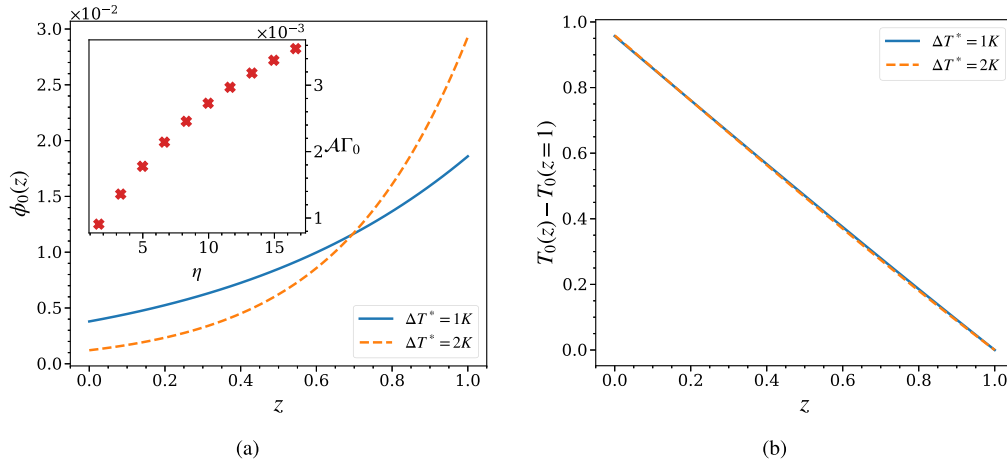


Fig. 2. Variation of the quiescent base state (a) concentration ϕ_0 and (b) temperature presented as $T_0(z) - T_0(z=1)$ with z across the layer at $B = 0.01$, $a = 7.47$, $\phi_m = 0.65$, $K_{ad} = 3.84$, $\mathcal{A} = 0.014$, $\Phi = 0.01$ and different values of the temperature difference ΔT^* . The inset of Fig. 2(a) represents the variation of the interfacial nanoparticle concentration Γ_0 with the Soret coefficient η .

substrate accordingly. Now, with this upper bound at hand, we seek the conditions to satisfy limitation (i). Satisfying the limitation (ii), for the fixed parameter set of $a = 7.47$, $\phi_m = 0.65$, $K_{ad} = 3.84$, $\mathcal{A} = 0.014$, $\rho_{np}^*/\mathcal{M}_{np}^* = 0.034$ along with $d_p^* = 50$ nm and the average particle concentration $\Phi = 0.01$ considered here, the temperature drop across the nanofluid layer can be considered up to $\Delta T^* \leq 10$ K, which corresponds in this case to $\phi_0(1) \leq 0.088$. Similarly, in terms of Φ , the upper bound of Φ should increase with a decrease in ΔT^* . For instance, for the same set of parameters, as defined above, with the temperature drop of $\Delta T^* = 1$ K, the average particle concentration is limited to $\Phi \leq 0.06$. Accordingly, an upper bound for ΔT^* imposes an upper bound for the admissible thermal Marangoni number M_T , which, for a nanofluid layer considered here with thickness $h_0^* = 10^{-6}$ m, must be limited by $M_T \leq 4.73$. The typical values of the physical parameters, both dimensional and non-dimensional, used in this paper are the same as in Gandhi et al. (2026b).

In nanofluids, one must also take into account the time scale related to the formation of particle aggregates. This provides another motivation for the restriction given by Eq. (27), which determines the upper bound of the nanoparticle concentration ϕ^* for a nanofluid to remain stable. It is related to the interaction potential Ψ^* of a polymer-stabilized suspension. We note that the nanoparticle volume fraction strongly influences the time scale of the doublet or the aggregate formation, which is given in Eq. (8.4.9) of Russel et al. (1989) and written in our own notation as

$$t_p^* = \frac{\pi \mu_{bf}^* d_p^{*3} W}{8 \phi^* K_B^* T^*}, \quad (28)$$

where W represents the stability ratio. The stability ratio corresponding to the energy barrier can be estimated via Eq. (8.5.4) in Russel et al. (1989)

$$W = W_\infty + 0.25 \left[\exp \left(\frac{\Psi_{\max}^*}{K_B^* T^*} \right) - 1 \right]. \quad (29)$$

Thus, for a sufficiently strong repulsive energy barrier, for instance, $\Psi_{\max}^*/K_B^* T^* \geq 21$, see Section 8.5 in Russel et al. (1989), and $W_\infty \approx 1$, the stability ratio yields $W \geq 3.3 \times 10^8$. Based on this, for a nanofluid with nanoparticles of diameter $d_p^* = 50$ nm and $\phi^* = 0.1$ (for instance) in a base fluid of viscosity $\mu_{bf}^* = 10^{-3}$ kg (ms) $^{-1}$ held at the temperature $T^* = 298$ K, the time scale for an aggregate formation is $t_p^* \geq 10.93$ h. Thus, in a regular experimental setting, the suspension will remain stable for more than 10 h. However, the flocculation time scale given by Eq. (28) is inversely proportional to the nanoparticle volume fraction ϕ^* . Note that for $\phi^* = 0.01$, the aggregation time significantly increases up to $t_p^* \approx 110$ h, i.e., up to 4–5 days. On the other hand, increasing

the nanoparticle concentration reduces the stability time scale and thus violates the underlying assumption of a stable nanofluid.

Additionally, the energy potential landscape contains the primary minimum associated with the tendency of nanoparticle attraction in the bulk fluid. In the case of polymeric stabilization of nanoparticle dispersion, namely, the polymer coating thickness on the surface of nanoparticles is tuned to weaken the interaction potential to the value $-\Psi_{\min}^*/K_B^* T^* \leq 2$. In such a case, the relaxation time for a nanofluid system to return to an equilibrium state is estimated by (Eq. (10.1.1) in Russel et al., 1989)

$$t_e^* = \frac{3\pi \mu_{bf}^* d_p^{*3}}{4K_B^* T^*} \exp \left(\frac{-\Psi_{\min}^*}{K_B^* T^*} \right). \quad (30)$$

Therefore, we find that under the weak attraction potential, the system with the same properties reversibly returns to its stable equilibrium state under the influence of Brownian diffusion in $t_e^* \approx 0.5$ ms.

To this end, we now turn to linear stability analysis of the system's base state given by Eq. (24).

3. Linear stability analysis

In this Section, we formulate the linear stability theory for the quiescent base state, $\mathbf{u}_0 = 0$, given by Eq. (24) for the flat interface at $z = h_0 = 1$. The dependent variables of the system are perturbed around the base states in the form

$$\begin{aligned} \mathbf{u} &= \mathbf{u}_0 + \bar{\mathbf{u}} = \bar{\mathbf{u}}, & p &= p_0 + \bar{p}, & T &= T_0 + \bar{T}, & \phi &= \phi_0 + \bar{\phi}, & \Gamma &= \Gamma_0 + \bar{\Gamma}, \\ h &= 1 + \bar{\zeta}, \end{aligned} \quad (31)$$

where the overbar decoration of the system variables denotes the perturbations of the respective fields. The equations mentioned above are then linearized. We note that the thermophysical properties of the nanofluid depend on the local particle concentration; hence, their values are also linearized around the base state concentration component ϕ_0

$$\begin{aligned} \rho_{nf} &= R_0 + \bar{R} = [\phi_m (\rho_{np} - 1) \phi_0 + 1] + [\phi_m (\rho_{np} - 1) \bar{\phi}], \\ (\rho c)_{nf} &= H_0 + \bar{H} = [\phi_m ((\rho c)_{np} - 1) \phi_0 + 1] + [\phi_m ((\rho c)_{np} - 1) \bar{\phi}], \\ K_{nf} &= K_0 + \bar{K} = (a \phi_m \phi_0 + 1) + \bar{\phi} a \phi_m, \\ \mu_{nf} &= M_0 + \bar{M} = \frac{1}{(\phi_0 - 1)^2} - \frac{2\bar{\phi}}{(\phi_0 - 1)^3}, \end{aligned} \quad (32)$$

where we denote a succinct symbol $X \equiv (R, H, K, M)$, representing the non-dimensional thermophysical properties, ρ_{nf} , $(\rho c)_{nf}$, K_{nf} , and μ_{nf} , respectively. The first term $X_0 \equiv (R_0, H_0, K_0, M_0)$ and the second

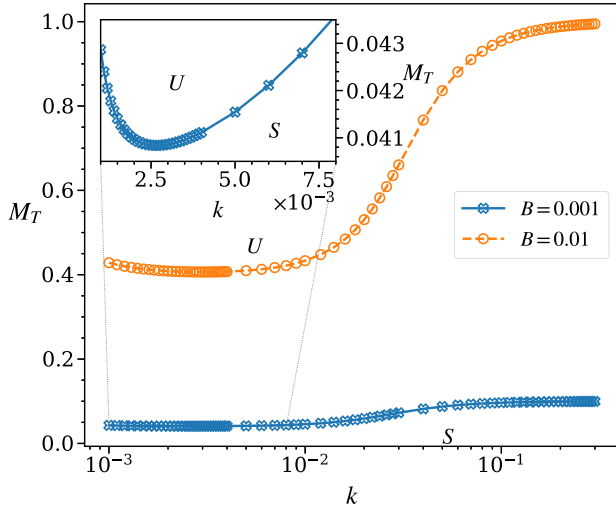


Fig. 3. Neutral curves $M_T(k)$ of the pure thermocapillary instability at $\Phi = 0.01$, $a = 7.47$, $\eta = 1.66$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, $G = 6.72 \times 10^{-6}$, $E = 0$ and $\Sigma_0 = 100$ for different values of the Biot number B . The inset shows a zoomed-in view of the neutral curve $M_T(k)$ for $B = 0.001$ near the instability threshold, depicting the onset of long-wave thermocapillary instability at a small Biot number. The symbols U and S denote the unstable and stable domains of the system, respectively.

term $\bar{X} \equiv (\bar{R}, \bar{M}, \bar{K}, \bar{H})$ in Eq. (32) denote the thermophysical properties evaluated at the z dependent base-state concentration and the disturbances, respectively. The problem at hand has the symmetry properties $x \leftrightarrow y$ and $u \leftrightarrow v$; therefore, we limit our investigation to the two-dimensional (2D) case in the $x - z$ plane. We eliminate the pressure from the problem by applying the curl $\equiv \nabla \times$ operator to the momentum conservation equations and by differentiating the normal stress balance boundary condition along the interface, followed by substituting there the pressure gradient components obtained from the momentum conservation equations.

We now introduce the normal perturbation modes to the system variables,

$$(\bar{u}, \bar{w}, \bar{T}, \bar{\phi}, \bar{\zeta}, \bar{\Gamma}; \bar{X})(x, z, t) = (u(z), w(z), \theta(z), \varphi(z), \zeta, \Gamma; X) \exp(ikx + \lambda t), \quad (33)$$

where u , w , θ , φ , ζ , Γ , and $X \equiv (R(z), M(z), K(z), H(z))$ are the amplitudes of the perturbations of the horizontal and vertical fluid velocity components, temperature, bulk nanoparticle concentration, interfacial deformation, surface particle concentration, and material properties (all expressed via φ), respectively. The stability of the system is determined with respect to the disturbance wavenumber $k \in \mathbb{R}$ via the growth rate $\lambda \in \mathbb{C}$ of the disturbances. In this representation, the negative and positive values of the real part of the growth rate λ correspond to the stability and instability regimes of the system, respectively. Further, we mention that X represents the material properties of the nanofluid, which are concentration dependent and do not introduce additional dependent variables following Eq. (32).

Substituting Eqs. (31) and (33) into Eqs. (15) and (16) yields the following eigenvalue problem (EVP) to determine the growth rate λ of the disturbance as a function of its wavenumber k for a specified parameter set:

$$R_0 (w' + iku) + \lambda PR + wR_0' = 0, \quad (34a)$$

$$\begin{aligned} & -iGkR + k^2 M_0 u' + 2k^2 u M_0' - ikw (k^2 M_0 + \lambda R_0 + M_0'') + ik M_0 w'' \\ & - u''' M_0 - 2u'' M_0' + \lambda R_0 u' - u' M_0' + \lambda u R_0' = 0, \end{aligned} \quad (34b)$$

$$-T_0'' K + \theta (k^2 K_0 + \lambda PH_0) + T_0' (wH_0 - K') - K_0 \theta'' - K_0' \theta' = 0, \quad (34c)$$

$$\begin{aligned} & \varphi (k^2 L - \eta LT_0'' + \lambda P) + \eta k^2 L \theta \phi_0 + \phi_0 (iku - \eta L \theta'' + w') \\ & + \phi_0' (w - \eta L \theta') - \eta LT_0' \varphi' - L \varphi'' = 0. \end{aligned} \quad (34d)$$

The resulting EVP contains the following boundary conditions: at the solid substrate $z = 0$:

$$\mathbf{u} = 0, \quad (a\phi_m \phi_0 + 1) \theta' + a\phi_m T_0' \varphi = 0, \quad \eta \phi_0 \theta' + \eta T_0' \varphi + \varphi' = 0. \quad (34e)$$

at the interface $z = 1$;

$$\zeta \lambda P - w = 0, \quad (34f)$$

$$\begin{aligned} & u (2k^2 M_0 + \lambda R_0) + i (-\zeta k P_0' + k M_0 w' - k M_0' w + i M_0 u'' + i M_0' u') \\ & + i \zeta k^3 [\Sigma_0 - M_T T_0 + E (K_\Sigma (\Gamma_0 (\omega_{np} - 1) + \omega_{np} \ln(1 - \Gamma_0)) \\ & + \phi_0 \phi_m (\ln(\Gamma_0) - \omega_{np} \ln(1 - \Gamma_0)))] = 0, \end{aligned} \quad (34g)$$

$$\begin{aligned} & M_0 (kw - iu') + k M_T (\zeta T_0' + \theta) - \frac{kE}{(\Gamma_0 - 1) \Gamma_0} \\ & \times \left[\Gamma \Gamma_0 K_\Sigma (\Gamma_0 (\omega_{np} - 1) + 1) \right. \\ & + \phi_m \left(-\Gamma \phi_0 + \Gamma_0^2 (\ln(\Gamma_0) - \omega_{np} \ln(1 - \Gamma_0)) (\zeta \phi_0' + \varphi) + \right. \\ & \left. \left. \Gamma_0 \left(-\zeta \ln(\Gamma_0) \phi_0' + \Gamma \phi_0 + \omega_{np} (\zeta \ln(1 - \Gamma_0) \phi_0' - \Gamma \phi_0) \right) \right) \right] \\ & + \varphi (\omega_{np} \ln(1 - \Gamma_0) - \ln(\Gamma_0)) \Big] = 0, \end{aligned} \quad (34h)$$

$$T_0' [a\phi_m (\zeta \phi_0' + \varphi) + B\zeta] + (a\phi_m \phi_0 + 1) \theta' + \zeta T_0'' (a\phi_m \phi_0 + 1) + B\theta = 0, \quad (34i)$$

$$\begin{aligned} & \zeta \phi_0'' + \eta \phi_0 (\theta' + \zeta T_0'') + \eta T_0' (\zeta \phi_0' + \varphi) + \varphi' \\ & = \mathcal{B}_D [K_{ad} ((\Gamma_0 - 1) \zeta \phi_0' + \Gamma \phi_0 + (\Gamma_0 - 1) \varphi) + \Gamma], \end{aligned} \quad (34j)$$

$$\begin{aligned} & P\lambda \Gamma + \Gamma k^2 L_s + i\Gamma_0 ku + \mathcal{B}_A [K_{ad} ((\Gamma_0 - 1) \zeta \phi_0' + \Gamma \phi_0 \\ & + (\Gamma_0 - 1) \varphi) + \Gamma] = 0. \end{aligned} \quad (34k)$$

We reiterate that R_0, M_0, K_0, H_0 represent the values of the thermophysical properties of the system given by Eq. (32), while R, M, K , and H are the amplitudes of their respective disturbances as functions of z . Furthermore, we note that, under non-equilibrium conditions, the independent perturbations of concentration fields, $\bar{\phi}$ and $\bar{\Gamma}$, violate the equilibrium relationship between $\phi_0(z = h)$ and Γ_0 given by Eq. (23c); however, interfacial kinetics and particle diffusion tend to equilibrate the system to its thermodynamically favorable state.

4. Results and discussion

The numerical solution of the obtained linear eigenvalue problem (EVP) follows the construction of the complex-valued Evans function by the Compound Matrix Method (Ng and Reid, 1979), whose roots determine the eigenvalues. A detailed description of the implementation of the EVP treatment in terms of only three independent variables w , θ , and φ is presented in our earlier publications (Gandhi et al., 2025, 2026b). We have favorably verified the accuracy of the numerical method by comparing it with the long-wave results obtained analytically (Gandhi et al., 2026b). We have also performed the accuracy check for our use of the compound matrix method, which is detailed in Table 1. We find that the leading eigenvalue of the eigenvalue problem (34) converges at a higher precision in terms of the number of digits used in the computations and the choice of the matching

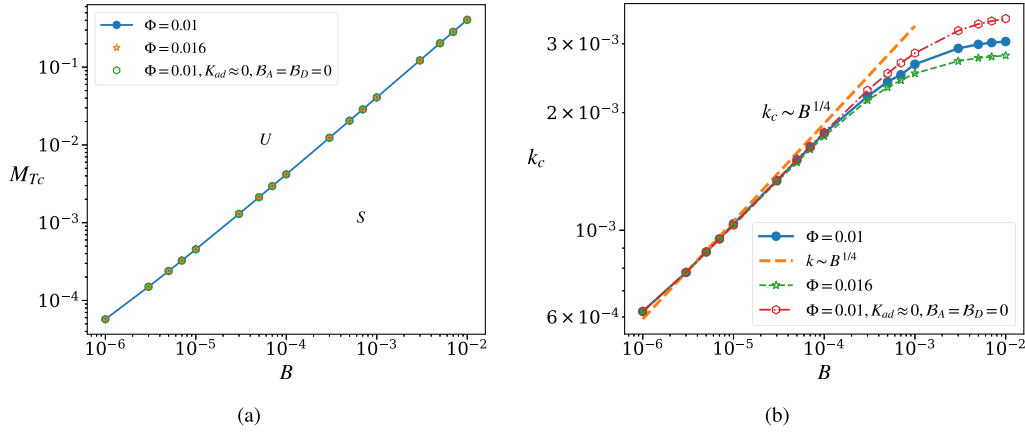


Fig. 4. Pure thermocapillary instability at $a = 7.47$, $\eta = 1.66$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, $G = 6.72 \times 10^{-6}$, $E = 0$ and $\Sigma_0 = 100$. (a) Variation of the critical thermal Marangoni number M_{Tc} for monotonic instability with the Biot number B . Note that the thermal Marangoni number exhibits a linear power-law fit with the Biot number, $M_{Tc} \approx 40.6235B$. The symbols U and S depict the unstable and stable domains of the system, respectively. (b) Variation of the critical wavenumber k_c with the Biot number B . The dashed line represents the data fit in the small Biot number limit, $k_c \approx 0.01874 B^{1/4}$, with a maximal deviation of 5.35% in the domain of $B \leq 10^{-4}$. The $\circ$, $\star$, and $\bigcirc$ points represent the instability thresholds at $\Phi = 0.01$, $\Phi = 0.016$, and $\Phi = 0.01$ with $K_{ad} \ll 1$ and $\mathcal{B}_A = \mathcal{B}_D = 0$, respectively.

Table 1

Numerical accuracy check for the evaluation of the leading eigenvalue of the eigenvalue problem (34) with respect to variation of the matching point and the working precision in the case of the thermocapillary instability at $B = 0.01$, $a = 7.47$, $\eta = 1.66$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, $G = 6.72 \times 10^{-6}$, $E = 0$ and $\Sigma_0 = 100$. Precision 16, 20 and 30 correspond to the use of 16, 20 and 30 digits in the computations, respectively.

Matching point	Eigenvalue at machine precision 16	Eigenvalue at precision 20	Eigenvalue at precision 30
$z = 1/4$	0.406497	0.406479	0.406481
$z = 1/2$	0.406538	0.406485	0.406481
$z = 3/4$	0.406554	0.406488	0.406479

point where the Evans function is evaluated. As seen in the example presented in Table 1, it remains almost insensitive to the choice of the matching point and the number of digits used in our numerical computations. Henceforth, we adopt the matching point at $z = 1/2$ within the integration domain and 30 digits of working precision for optimal accuracy.

The linear EVP (34) at hand is presented in a ‘compressible’ form due to the dependence of density on local nanoparticle concentration, which in turn depends on space and time. Nevertheless, for moderately dense nanosuspension $\Phi \leq 0.1$ and small Lewis numbers $L \in (10^{-5}, 10^{-3})$, the system can be treated in its ‘incompressible’ approximation, which was validated by direct comparison between the solutions of the full EVP in its compressible and incompressible forms (Gandhi et al., 2025). Henceforth, we solve the eigenvalue problem (34) under the assumption of an incompressible fluid system.

4.1. Thermocapillary instability

A layer of a pure Newtonian fluid or a mixture, such as a binary mixture or a nanofluid, heated on its solid substrate, undergoes thermocapillary instability due to a decrease in surface tension with the growth of temperature. This thermocapillary or Marangoni instability may emerge in both the short- and long-wave range (Davis, 1987; VanHook et al., 1997; Oron et al., 1997; Schatz and Neitzel, 2001; Gandhi et al., 2025). We emphasize that in the case of a nanofluid, the emerging mode of thermocapillary instability strongly depends on the values of the system parameters and the interplay between different instability mechanisms (Gandhi et al., 2025).

Fig. 3 illustrates the neutral curves $M_T(k)$ for the onset of long-wave monotonic pure thermocapillary instability for different Biot numbers B . As always, we note that with an increase in the Biot number, the system stabilizes by losing some of the heat to the atmosphere, and therefore, it requires a higher temperature drop across the layer or a higher heat flux supply at the substrate to compensate for the heat loss to the atmosphere, and thus a higher value of the critical thermal Marangoni number M_{Tc} to destabilize the system (Davis, 1987). We also find that the critical thermal Marangoni number M_{Tc} linearly varies with the Biot number, $M_{Tc} \sim B$. For small values of B , the critical wavenumber k_c exhibits the well-known power-law variation $k_c \sim B^{1/4}$, and saturates to an almost constant value for larger values of B , as shown in Fig. 4.

Fig. 5(a) illustrates the stabilization of the system in terms of the thermal Marangoni number M_{Tc} , which exhibits a linear increase with the dimensionless surface tension number Σ_0 , $M_{Tc} \sim \Sigma_0$. This follows from the fact that surface tension effectively reduces interfacial deformation, namely the distance between the substrate and the troughs, in turn delaying the onset of thermocapillary instability by raising the threshold M_{Tc} (Davis, 1983; Oron et al., 1997). Consequently, the surface tension causes an increase in the length of the emerging cells and therefore reduces the critical wavenumber. Furthermore, Fig. 5(b) presents the variation of the critical wavenumber k_c with the dimensionless surface tension number Σ_0 following two different scalings corresponding to the long-wave and short-wave domains. In the regime of moderate values of Σ_0 , the critical wavenumber follows the long-wave scale $k_c \sim \Sigma_0^{-1/2}$, whereas in that of small values of Σ_0 , the critical wavenumber follows the short-wave regime scaling $k_c \sim \Sigma_0^{-1/4}$. We note that the variation of critical values of the Marangoni number and the wavenumber with Σ_0 and B remains qualitatively consistent with the variation in the average particle concentration Φ , with and without interfacial kinetics, as shown in Figs. 4 and 5, respectively. This implies that the presence of the interfacial kinetics does not introduce new long-wave scales.

The presence of gravity plays a substantial role in the case of long-wave deformational instability by dampening interfacial perturbations. Fig. 6 illustrates the variation of the critical value of the thermal Marangoni number M_T with the modified Galileo number G . We note that G weakly stabilizes the thermocapillary instability for the given system. However, we recall that for a nanofluid layer with heavy nanoparticles $\rho_{np} = 4$, there exists an inherent solutal buoyancy instability (Gandhi et al., 2025), although in the case of a thin nanofluid layer $h_0^* \approx 10^{-6}$ m, the solutal buoyancy instability does not emerge.

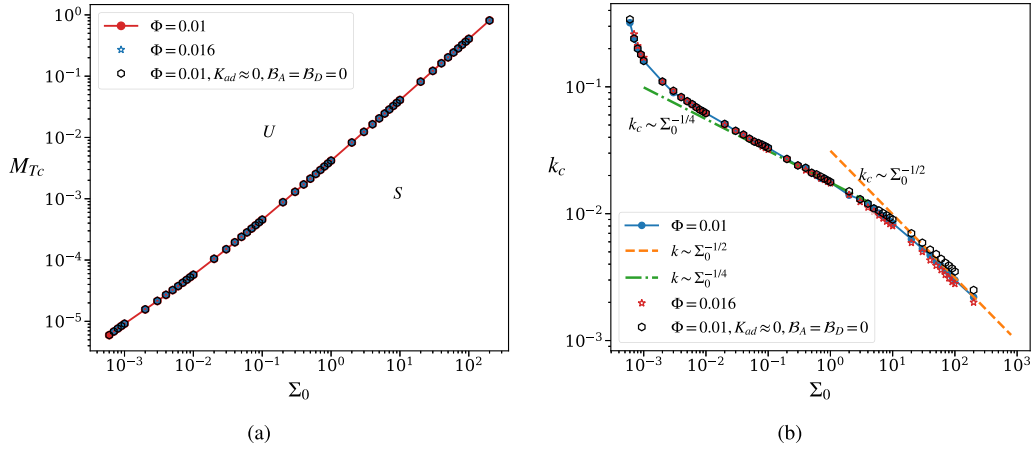


Fig. 5. Pure thermocapillary instability at $a = 7.47$, $\eta = 1.66$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_Y = 2.11 \times 10^{-3}$, $G = 6.72 \times 10^{-6}$, $E = 0$ and $B = 0.01$. The $\circ$, $\star$, and $\circ$ points represent thresholds at $\Phi = 0.01$, $\Phi = 0.016$, and $\Phi = 0.01$ with $K_{ad} \ll 1$ and $\mathcal{B}_A = \mathcal{B}_D = 0$, respectively. (a) Variation of the critical thermal Marangoni number M_{Tc} of the monotonic instability with the dimensionless surface tension number Σ_0 . Note that the thermal Marangoni number exhibits a linear power-law fit $M_T \approx 0.004 \Sigma_0$. The symbols U and S represent the unstable and stable domains of the system, respectively. (b) Variation of the critical wavenumber k_c with Σ_0 . Note that the system exhibits the emergence of two power-law scalings in two different ranges of Σ_0 : $k_c \approx 0.0313 \Sigma_0^{-1/2}$, with a maximal deviation of 10% in the domain of $20 \leq \Sigma_0 \leq 400$, for large Σ_0 and $k_c \approx 0.0176 \Sigma_0^{-1/4}$, with a maximal deviation of 16.67% in the domain of $0.003 \leq \Sigma_0 \leq 4$, for small Σ_0 .

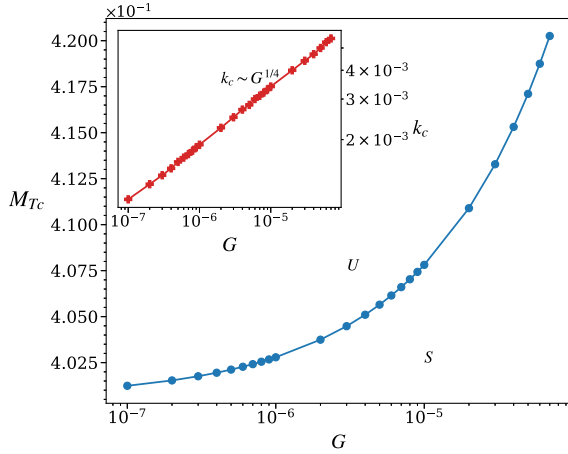


Fig. 6. Variation of the critical thermal Marangoni number for the monotonic thermocapillary instability threshold M_{Tc} with the modified Galileo number G at $a = 7.47$, $\eta = 1.66$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_Y = 2.11 \times 10^{-3}$, $\Sigma_0 = 100$, $E = 0$, $\Phi = 0.01$, and $B = 0.01$. The inset shows the variation of the critical wavenumber k_c with G , $k_c \approx 0.06013 G^{1/4}$. The symbols U and S denote the unstable and stable domains of instability, respectively.

Moreover, we find that the critical wavenumber k_c increases with the modified Galileo number G , as shown in the inset of Fig. 6, following the law $k_c \sim G^{1/4}$. We further stress that the analytical expression of the thermocapillary threshold found for the toy problem in Gandhi et al. (2026b) qualitatively predicts the long-wave thermocapillary threshold for the variation of M_{Tc} with the Biot number B , the dimensionless surface tension number Σ_0 , and the modified Galileo number G . We emphasize that, however small the values of the modified Galileo number G are, the latter is necessary to materialize the approximate analytical structure for the given scales.

The emergence of thermocapillary instability in the present setting is qualitatively different from that discussed in Gandhi et al. (2026b) in the case of a heated layer of a nanofluid with no particle kinetics due to the presence of nanoparticle kinetics at the interface and a different type of surface tension variation, which in the present case depends on both the surface and bulk concentrations of the particles. Therefore,

the underlying physical mechanisms responsible for the appearance of thermocapillary instability are different and will be discussed next.

We now examine the mechanism of the underlying long-wave thermocapillary instability. The emergence of thermocapillary instability is investigated near criticality, i.e., at the critical wavenumber $k_c = 0.00304$, where the growth rate is the highest for the given control parameter, namely the thermal Marangoni number $M_T = 0.41$, which is slightly higher than the threshold value $M_{Tc} = 0.4064$. Figs. 7(a), 7(b), 7(c) and 7(d) present the normalized eigenfunctions for the components of perturbed temperature $\tilde{T}(x, z)$, bulk nanoparticle concentration $\tilde{\phi}(x, z)$, both superimposed with the velocity field $\tilde{u}(x, z)$, the interfacial concentration $\tilde{F}(x)$, and the interface deformation $\tilde{\xi}(x)$, respectively. We recall that for the given surface tension model (10), the surface tension linearly decreases with the interfacial temperature; therefore, if due to an infinitesimal perturbation, a fluid packet near the substrate is displaced towards the interface, it creates a hot spot at the liquid–air interface. For a higher surface tension gradient, the viscous stress at the interface is insufficient to counteract the flow due to the thermocapillary stress; subsequently, due to mass conservation, the system exhibits an upwelling flow beneath this hot spot. However, the structure of the eigenfunctions in Figs. 7(a) and 7(b) seems to contradict the conventional structure of thermocapillary flow; although they are physically consistent, considering the presence of the positive Soret effect.

In particular, in the case of the long-wave Marangoni instability, interfacial deformation plays a vital role in the onset of instability and the evolution of long-wave modes (VanHook et al., 1997; Or et al., 1999; Schatz and Neitzel, 2001). The interfacial deformation shown in Fig. 7(d) being dominant over the rest of the eigenfunctions along with the temperature eigenfunction $\tilde{T}(x, z)$, both of $O(1)$, is responsible via the induced temperature for the emergence of the Marangoni flow. In such a situation, not the temperature at the equilibrium surface $z = 1$ is relevant, but the variation of the temperature along the deformed interface $z = 1 + \tilde{\xi}(x)$, which must be accounted for to consider the onset of thermocapillary instability.

Fig. 7(e) displays the distribution of the total temperature along the deformed interface. We note that the system exhibits a higher temperature at the trough of the interface, where the surface tension is lower compared to that at the crest of the interface. Therefore, as described earlier, the flow emanates from the trough towards the crest. Fig. 7(c) shows the depletion of particles in the valley of the deformed interface, which occurs due to the higher temperature there and the Soret effect driving the particles towards the cooler domain.

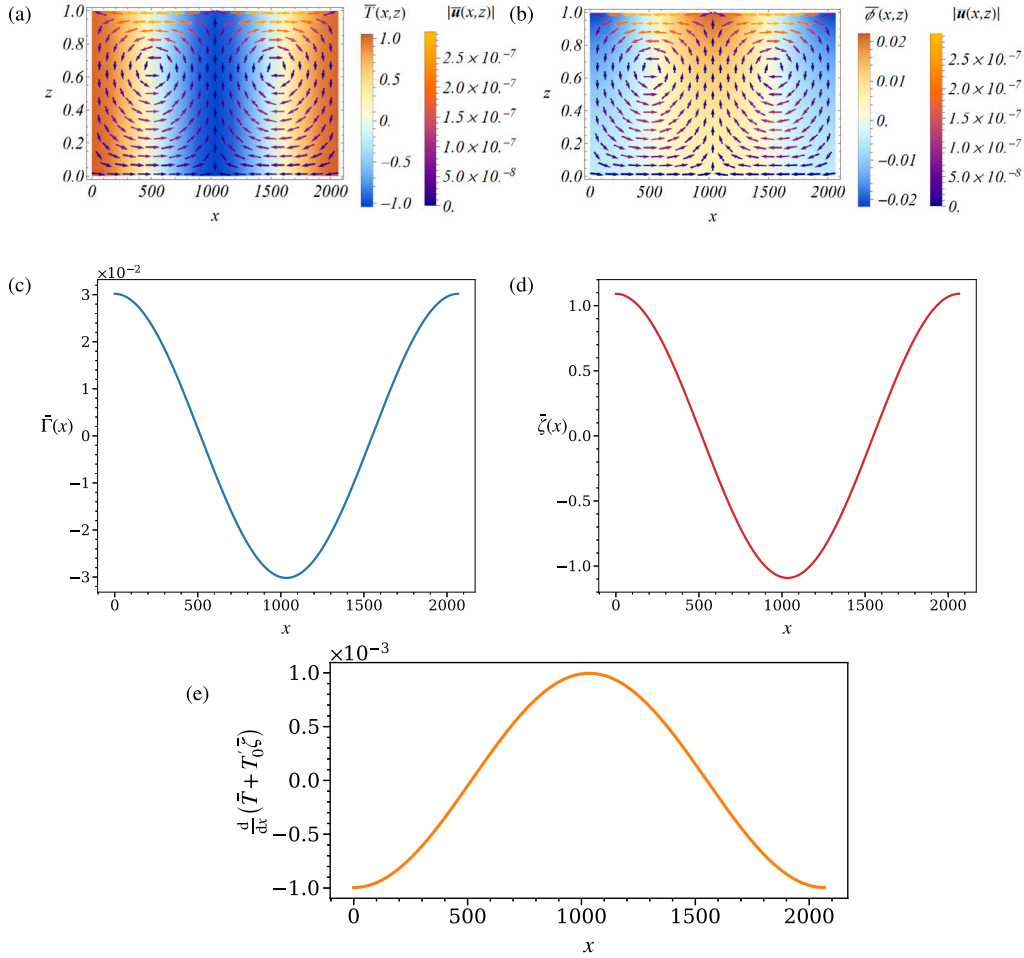


Fig. 7. Normalized dimensionless eigenfunctions of the EVP (34) corresponding to the onset of long-wave pure thermocapillary instability for the Marangoni number $M_T = 0.41$ and the wavenumber $k_c = 0.00304$ at $B = 0.01$, $\Phi = 0.01$, $G = 6.7 \times 10^{-6}$, $\Sigma_0 = 100$, $\eta = 1.66$, $L = L_s = 6 \times 10^{-3}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.72$, $K_{ad} = 3.84$, $E = 0$, $P = 7$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, and $\lambda = 3.36 \times 10^{-12}$. (a) and (b): the eigenfunctions $\tilde{T}(x, z)$ and $\tilde{\phi}(x, z)$, respectively, superimposed with the velocity vector field $\tilde{\mathbf{u}}(x, z)$. (c) and (d) the disturbance of the interfacial nanoparticle concentration $\tilde{F}(x)$ and the deformation $\tilde{\zeta}(x)$, respectively. Note the emergence of long-wave deformational thermocapillary instability. (e) The directional derivative along the interface of the total interfacial temperature $\frac{d}{dx}(\tilde{T} + T_0 \tilde{\zeta})$, which is proportional to the total thermocapillary stress responsible for the emergence of the long-wave deformational thermocapillary instability. The eigenfunctions are normalized to make the maximum amplitude of the dominant eigenfunction of the temperature field equal to one. Note that the prime denotes differentiation with respect to z .

4.2. Solutocapillary instability

We recall that the base state concentration profile, shown in Fig. 2(a), displays an unstable density stratification due to the positive value of the Soret coefficient. Therefore, in the case of a linear decrease of surface tension with nanoparticle concentration in the absence of interfacial particle kinetics, solutocapillarity stabilizes the system (Gandhi et al., 2025). However, in the presence of interfacial particle kinetics, non-uniformity of surface tension can also emerge due to the advection of adsorbed interfacial nanoparticles/surfactants or due to an inhomogeneous adsorption of nanoparticles from the bulk (Manikantan and Squires, 2020). Furthermore, irrespective of the direction of heating/cooling, spatial redistribution of particles on the surface due to adsorption/desorption kinetics may cause inhomogeneity of the surface particle concentration, hence a possibility of the emergence of the solutocapillary instability (Shmyrov et al., 2019). We further emphasize that the nonmonotonic variation of surface tension with the nanoparticle concentration also promotes the onset of *anomalous* solutocapillary instability (Priede et al., 1999; Braverman et al., 2000; Gandhi et al., 2026a). In all of the scenarios mentioned above, particularly for a complex system like the one considered here, the collective action of

various inherent physical mechanisms determines the wavelength of the emerging instability.

Fig. 8(a) presents the neutral curves $E(k)$ in the case of the long-wave monotonic purely solutocapillary instability for several values of the Biot number B . We note that an increase in the Biot number causes an increase in the critical Elasticity number E_c , and that the solutocapillary instability remains long-wave, as shown in the inset to Fig. 8(b). With an additional increase in B , the critical Elasticity number E_c saturates, as follows from Fig. 8(b). Similarly, in the domain of small Biot number B , the critical wave number k_c follows the power-law scale $k_c \sim B^{1/4}$. To understand the variation of the threshold value of the Elasticity number E with different parameters, we explore the physical mechanism behind the onset of long-wave solutocapillary instability.

We now address the structure of the long-wave mode emerging at the threshold of the solutocapillary instability by presenting the normalized eigenfunctions near criticality. Figs. 9(a) to 9(d) present normalized eigenfunctions for the components of the perturbed temperature $\tilde{T}(x, z)$, the concentration of bulk nanoparticles $\tilde{\phi}(x, z)$, both superimposed on the velocity field $\tilde{\mathbf{u}}(x, z)$, the interfacial concentration $\tilde{F}(x)$, and the interface deformation $\tilde{\zeta}(x)$, respectively. Note that in the present setting, thermocapillarity is absent, $M_T = 0$. We recall

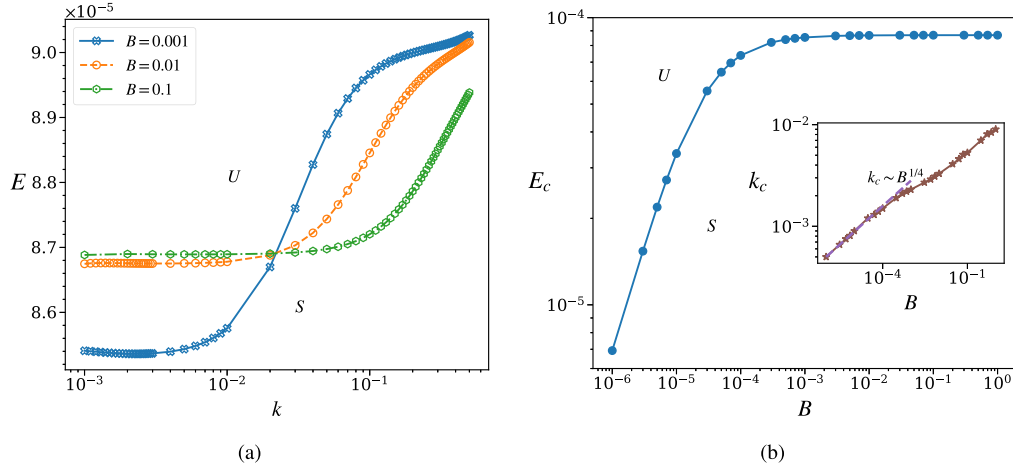


Fig. 8. Monotonic purely solutocapillary instability at $\Phi = 0.01$, $a = 7.47$, $G = 6.72 \times 10^{-6}$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, $\Sigma_0 = 100$, $\eta = 1.66$ and $M_T = 0$ (a) Neutral curves $E(k)$ for different Biot numbers. The U and S symbols depict the unstable and stable domains of the instability, respectively. (b) Variation of the critical Elasticity number E_c with B . Inset: Variation of the critical wavenumber k_c with the Biot number B . The dashed line represents the long-wave power-law data fit, $k \sim B^{1/4}$ with the proportionality constant 0.01581.

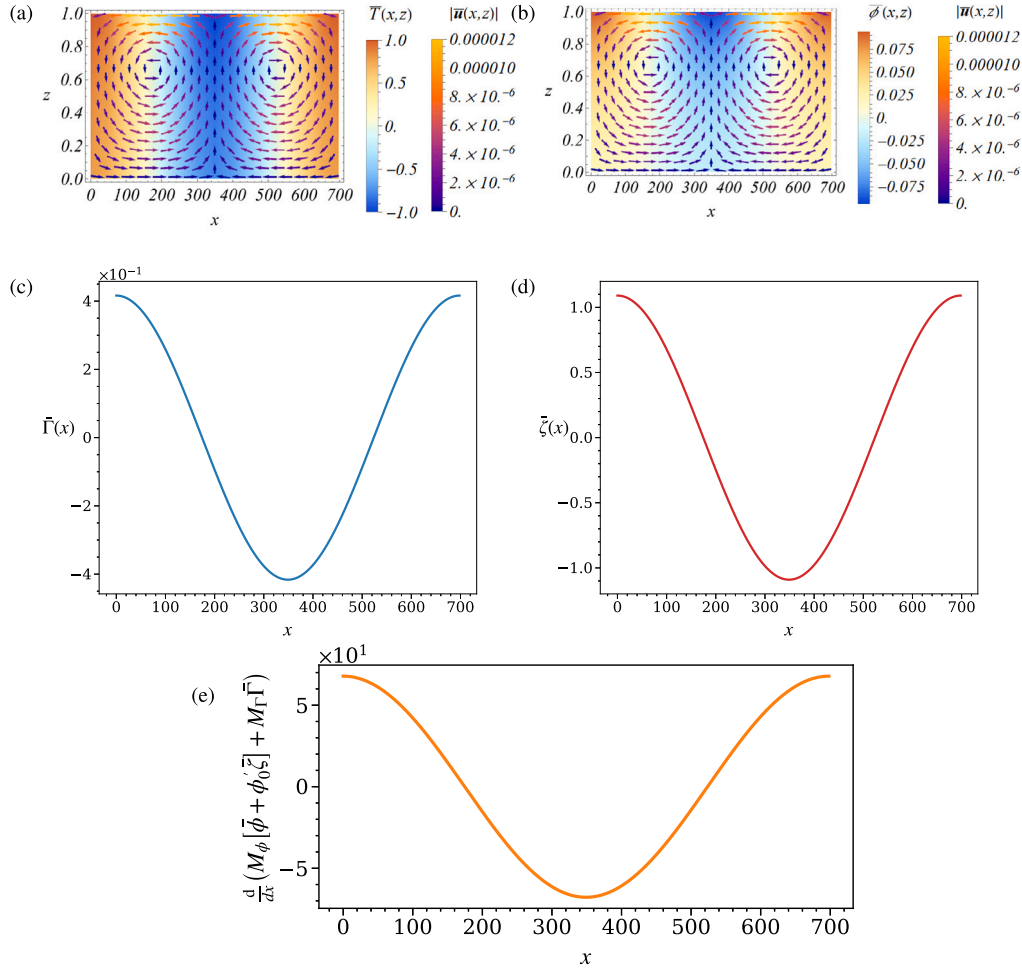


Fig. 9. Normalized dimensionless eigenfunctions of the EVP (34) corresponding to the onset of a pure long-wave deformational solutocapillary instability for the critical wavenumber $k_c = 0.009$ at $B = 1$, $\Phi = 0.01$, $G = 6.7 \times 10^{-6}$, $\Sigma_0 = 100$, $\eta = 1.66$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.72$, $K_{ad} = 3.84$, $P = 7$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, $E = 8.71 \times 10^{-5}$, $M_T = 0$, and $\lambda = 1.7529 \times 10^{-12}$. (a) and (b): the eigenfunctions $\tilde{T}(x, z)$ and $\tilde{\phi}(x, z)$, respectively, superimposed with the velocity vector field $\tilde{\mathbf{u}}(x, z)$. (c) and (d): the disturbances of the interfacial nanoparticle concentration $\tilde{\Gamma}(x)$ and the interfacial deformation $\tilde{\zeta}(x)$, respectively. (e): The directional derivative along the interface of the total interfacial nanoparticle concentration $\tilde{\tau}_{xz} \sim \frac{d}{dx} (M_\phi [\tilde{\phi} + \phi'_0 \tilde{\zeta}] + M_T \tilde{\Gamma})$, which is proportional to the total solutocapillary stress responsible for the emergence of the long-wave deformational solutocapillary instability. The eigenfunctions are normalized to make the maximum amplitude of the dominant eigenfunction of the temperature field equal to one.

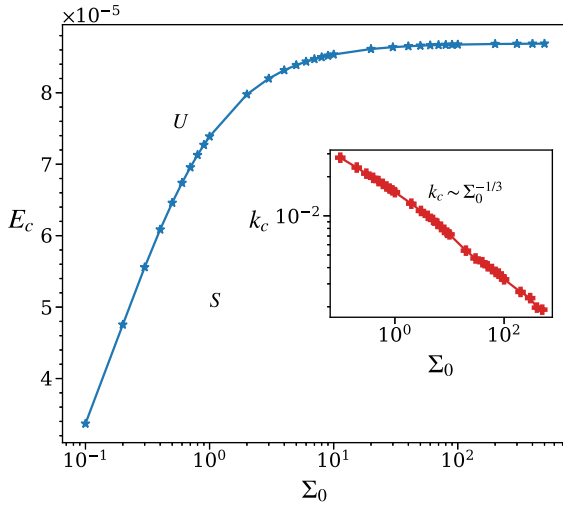


Fig. 10. Variation of the critical Elasticity number E_c with the dimensionless surface tension number Σ_0 at $\Phi = 0.01$, $B = 0.01$, $a = 7.47$, $G = 6.72 \times 10^{-6}$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_S = 2.11 \times 10^{-3}$, $\eta = 1.66$ and $M_T = 0$. The inset: the variation of the critical wavenumber k_c with Σ_0 at $\Phi = 0.01$, which follows the power-law scale $k_c \sim \Sigma_0^{-1/3}$ with the proportionality constant 0.015085. Note that gravity weakly affects the threshold of the solutocapillary instability; however, $k_c \sim G^{1/4}$.

that in the case of nanoparticle dispersion $\text{Al}_2\text{O}_3\text{-W}$, the surface tension increases with the nanoparticle concentration (Machrafi, 2022; Gandhi et al., 2026b); therefore, the anomalous solutocapillary instability emerges by dragging the fluid from the spot at the interface with the low particle concentration to that with the high particle concentration, as displayed in Fig. 9(b).

We also note that the interfacial deformation, being of order $O(1)$ as shown in Fig. 9(d), dominates the contributions of other perturbed fields. Thence, similar to the long-wave thermocapillary instability discussed in the previous section, we analyze the variation of the solutocapillary stress along the deformed interface. The variation of the total solutocapillary stress, which is proportional to the critical Elasticity number, is given by

$$\bar{\tau}_{xz} \sim \frac{d}{dx} (M_\phi [\bar{\phi} + \phi'_0 \bar{\zeta}] + M_T \bar{F}). \quad (35)$$

is shown in Fig. 9(e). According to this variation, the solutal Marangoni stress is directed from the trough where $\bar{\tau}_{xz}$ is the lowest, thus where the surface tension is the lowest, towards the crests with the highest interfacial concentration, thus where the surface tension is the highest. We also find that the contributions of the terms with $\bar{\zeta}(x)$ and $\bar{F}(x)$ to the total solutocapillary stress $\bar{\tau}_{xz}$ are dominant for the emergence of an anomalous solutocapillary instability.

Fig. 10 presents the variation of the critical Elasticity number E_c with the dimensionless surface tension number Σ_0 . We note that with an increase in the dimensionless surface tension number Σ_0 , the system is significantly stabilized since the critical value of E increases, since an increase in the surface tension promotes damping of interfacial perturbations. Naturally, we observe that the critical wavenumber k_c decreases with Σ_0 and exhibits a power-law fit $k_c \sim \Sigma_0^{-1/3}$. On the other hand, we find that the modified Galileo number G has a minor effect on the critical Elasticity number (not shown here). Similar to the case of a pure thermocapillary instability, the critical wavenumber follows the scaling $k_c \sim G^{1/4}$.

We find that the Soret effect, whose strength is described by η tends to promote the solutocapillary instability. This fact is expressed by a reduction of the critical Elasticity number E_c , as shown in Fig. 11(a). Fig. 2(a) helps to clarify the reason for this: an increase in η

is associated with an enhancement of the Soret effect and thus leads to an increase in both interfacial nanoparticle concentration and bulk nanoparticle concentration near the surface.

Similarly, an increase in the average bulk nanoparticle concentration Φ lowers the critical Elasticity number E_c and therefore destabilizes the system. This is consistent with the dependence of M_ϕ and M_T on the equilibrium particle concentration shown in Gandhi et al. (2026b), which leads, with an increase of the particle concentration near the surface, to strengthening the solutocapillarity and, therefore, to lowering the instability threshold, while the critical wavenumber k_c remains almost constant $k_c \approx 0.0032$. Importantly, we stress that for a stronger Soret effect (larger η), the solutocapillary instability is long-wave, similar to what is shown in Fig. 8(a). However, with a decrease in η (weakening of the Soret effect), the solutocapillary instability becomes a short-wave instability, as shown in Fig. 11(b).

It is interesting to note that the thermal conductivity stratification parameter a reduces the base state component of the bulk particle concentration ϕ_0 at the surface $z = 1$ and the interfacial particle concentration Γ_0 , as shown in Fig. 11(d). Hence, an increase in a promotes a weakening of the solutocapillary instability observed through an increase in the critical value of the Elasticity number E shown in Fig. 11(c). This consequently implies that the particle concentration fluctuations at the surface are inhibited. Fig. 11(c) also displays a gradual decrease in the critical wavenumber k_c with a .

4.3. Combined thermo-solutocapillary instability

In this subsection, we briefly discuss the combined effect of thermocapillary and anomalous solutocapillary instabilities. Fig. 12(a) presents the threshold of the combined thermo-solutocapillary instability in the plane of $E_c/E_{c0} - M_{Tc}/M_{Tc0}$ where M_{Tc0} and E_{c0} are the critical values of the thermal Marangoni and the Elasticity numbers for the pure thermocapillary and solutocapillary instabilities, respectively. We note that the temperature and concentration components of the surface tension both destabilize the system and lead to its instability, since the normalized thermal Marangoni number M_{Tc}/M_{Tc0} decreases with an increase in the value of the normalized Elasticity number E_c/E_{c0} . Interestingly, the $E_c/E_{c0} - M_{Tc}/M_{Tc0}$ plane reveals a strong coupling between both capillary-driven instabilities. The threshold of the combined thermo-solutocapillary instability exhibits a slight curvature for intermediate values of the normalized Elasticity numbers. Fig. 12(b) displays the non-monotonic variation of the critical wavenumber k_c with the Elasticity number E_c along the critical curve displayed in Fig. 12(a), which shows that the instability is long-wave. The critical wavenumber k_c decreases almost linearly with E_c for lower values of E_c and subsequently increases rapidly towards the value of E_{c0} corresponding to the pure solutocapillary instability.

In the limiting case of a nanofluid layer (Jensen and Grotberg, 1992; Quéré, 1999; Delacotte et al., 2012; Shklyaev and Nepomnyashchy, 2013; Manikantan and Squires, 2020; Mayer et al., 2024), with slow adsorption/desorption kinetics $\mathcal{B}_A$ and $\mathcal{B}_D \rightarrow 0$ but with a finite K_{ad} , insoluble interfacial nanoparticle concentration affects short capillary waves, for more detail see Levich (1962), Manikantan and Squires (2020), and Gandhi et al. (2026b). However, we find that the thermo-solutocapillary instability considered here remains long-wave also in the absence of interfacial particle kinetics.

5. Summary and conclusions

The present paper highlights the influence of interfacial adsorption/desorption kinetics and the non-monotonic variation of surface tension with nanoparticle concentration on the onset of thermosolutal instability in a moderately dense nanofluid layer. Our nanofluid system and mathematical model follow our previous investigation (Gandhi et al., 2026b); however, in the current setting, we consider a thin

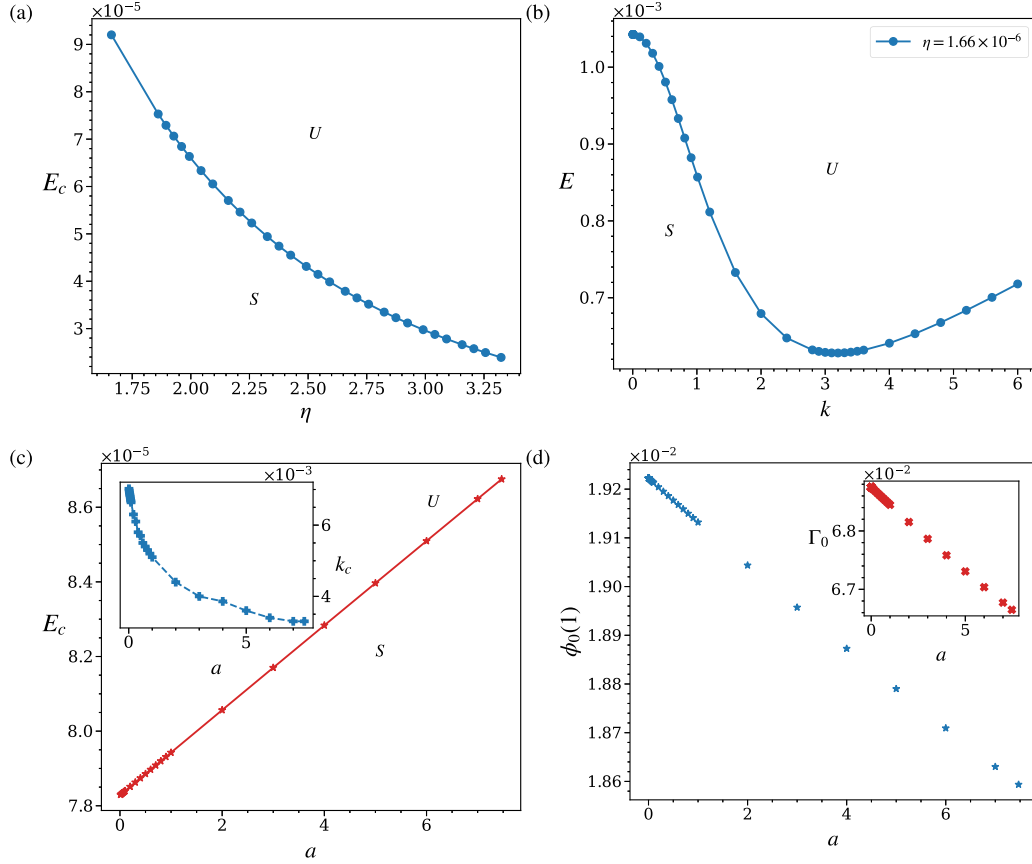


Fig. 11. Pure monotonic solutocapillary instability at $\Phi = 0.01$, $B = 0.01$, $G = 6.72 \times 10^{-6}$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, $\Sigma_0 = 100$, and $M_T = 0$. (a): variation of the critical Elasticity number E_c with the Soret coefficient η at $a = 7.47$. Here, the critical wavenumber k_c remains almost constant, $k_c \approx 0.0032$. (b): Neutral curve $E(k)$ in the case of a small Soret coefficient $\eta = 1.66 \times 10^{-6}$. (c): Variation of the critical Elasticity number with thermal conductivity stratification parameter a at $\eta = 1.66$. The inset: variation of the critical wavenumber k_c with a at $\eta = 1.66$. (d) Variation of the base state component of the bulk nanoparticle concentration ϕ_0 at $z = 1$ with the thermal conductivity stratification parameter a at $\eta = 1.66$ and $\Phi = 0.01$. The inset: variation of the base state component of the interfacial nanoparticle concentration Γ_0 with a . In panels (a), (b), and (c), the symbols U and S represent the unstable and stable domains of the system.

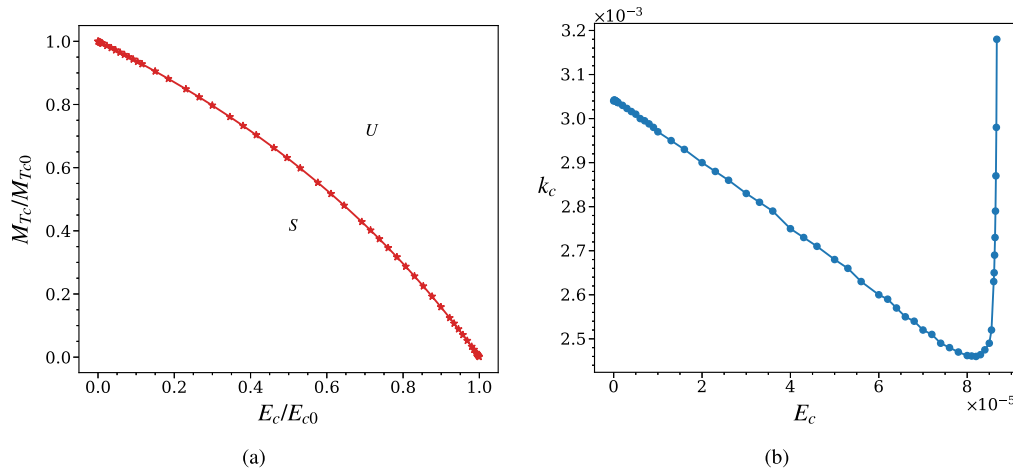


Fig. 12. (a) Threshold of the combined thermo-solutocapillary instability displayed in the plane $E_c/E_{c0} - M_{Tc}/M_{Tc0}$ at $\Phi = 0.01$, $G = 6.72 \times 10^{-6}$, $L = L_s = 6 \times 10^{-5}$, $\mathcal{B}_A = 0.01578$, $\mathcal{B}_D = 6.7236$, $K_{ad} = 3.84$, $\phi_m = 0.65$, $\mathcal{A} = 0.014$, $\omega_{np} = 3.23 \times 10^3$, $K_\Sigma = 2.11 \times 10^{-3}$, $B = 0.01$, $\Sigma_0 = 100$, $\eta = 1.66$, and $a = 7.47$. The threshold values of the thermal Marangoni number M_{Tc} and the Elasticity number E_c are normalized with their critical values corresponding to the purely thermocapillary and solutocapillary instabilities, $M_{Tc0} = 0.4065$ and $E_{c0} = 8.67 \times 10^{-5}$, respectively. The U and S symbols denote the monotonic unstable and stable domains of the system, respectively. (b) Variation of the critical wavenumber k_c with the Elasticity number E along the critical curve shown in Fig. 12(a). In the limit of slow adsorption and desorption, $\mathcal{B}_A$ and $\mathcal{B}_D \rightarrow 0$ and finite K_{ad} , a qualitatively similar variation in the $E_c - M_{Tc}$ plane is observed.

nanofluid layer heated by a constant heat flux at the poorly conducting solid substrate and exposed to the ambient atmosphere at its deformable interface in the presence of gravity and interfacial kinetics expressed by adsorption and desorption. Our mathematical model considers thermophysical properties of the nanofluid that depend on the local bulk particle concentration with the total nanoparticle mass flux containing the components of the Brownian diffusion and the Soret effect.

We reveal the emergence of long-wave thermocapillary instability and note that the system stabilizes with an increase in the Biot number B , the dimensionless surface tension number Σ_0 , and the modified Galileo number G . The critical value of the Marangoni number M_{Tc} varies linearly with both B and Σ_0 ; however, an increase in G imparts a weak stabilization in the parameter range considered here. In the long-wave domain, the critical wavenumber k_c follows power-law scalings $k_c \sim B^{1/4}$, $k_c \sim \Sigma_0^{-1/2}$, and $k_c \sim G^{1/4}$. We corroborate our findings by identifying the inherent long-wave deformational mode of the thermocapillary instability through appropriate eigenfunctions.

We further note that in the presence of interfacial particle kinetics and a positive surface-tension gradient with particle concentration, an anomalous monotonic solutocapillary instability emerges in the long-wave domain. Similar to the log-wave thermocapillary instability, we present the eigenfunctions corresponding to the emergence of the solutocapillary instability, which consequently clarifies the role of interfacial particle concentration and the deformation in the onset of the long-wave solutocapillary instability. Likewise, we find that an increase in the Biot number and the dimensionless surface tension number stabilizes the solutocapillary instability by increasing the critical Elasticity number; additionally, the critical wavenumber follows the long-wave scalings, $k_c \sim B^{1/4}$ and $k_c \sim \Sigma_0^{-1/3}$. Moreover, we find that an increase in the average particle concentration Φ and the Soret coefficient η significantly amplifies the amplitude of the interfacial perturbation due to the surface particle concentration and the interfacial deformation. Therefore, we note a substantial drop in the critical Elasticity number with an increase in Φ and η . However, interestingly, we reveal that an increase in the thermal conductivity stratification parameter a causes a decrease in the interfacial particle concentration; consequently, its increase stabilizes the system in terms of the solutocapillary instability.

We distinguish between the results obtained in the presence of interfacial particle kinetics here and those presented in Gandhi et al. (2025), where kinetics were not taken into account. A significant reduction in the modified Galileo number G is found to promote the emergence of new long-wave instability regimes, which are complementary to the long-wave regimes in Gandhi et al. (2025). In particular, with no particle kinetics at the layer interface, in the case of heating at the substrate, the solutocapillarity is stabilizing; however, in the presence of interfacial kinetics, the solutocapillary instability emerges. We reveal that a strong coupling between the temperature and concentration fields propels the system towards the emergence of the solutocapillary instability in the long-wave domain driven by the Soret effect. An intricate role of thermal effects on the threshold of the solutocapillary instability is identified, and the possibility of the emergence of the long-wave instability similar to those presented in Gandhi et al. (2026b,a) is revealed. To conclude, we emphasize that our investigation reveals a plethora of scales in the long-wave instabilities in the presence of interfacial kinetics. These scales are crucial for developing and investigating the nonlinear dynamics of the long-wave thermo-solutocapillary instability. The findings of this paper address the scientific questions outlined in the introduction section and provide a new direction and understanding of complex systems in emerging technologies, such as microfluidics, chemical processing, colloidal crystals, and others.

CRediT authorship contribution statement

Raj Gandhi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Formal analysis. **Alexander Nepomnyashchy:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Funding acquisition, Formal

analysis, Conceptualization. **Alexander Oron:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was partially supported by the grant from the European Union's Horizon 2020 research and innovation program under the Marie Skłodowska-Curie grant agreement number 955612 (NanoPaInt). R. G. was partially supported by the fellowship granted to him by the Irwin & Joan Jacobs Graduate School at the Technion - Israel Institute of Technology.

Data availability

Data will be made available on request.

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